

BandHiC

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Welcome to the BandHiC documentation!

This documentation provides a comprehensive guide to the BandHiC library, including its features, installation instructions, and API reference. It is designed to help you efficiently store and manipulate Hi-C contact matrices in a banded form, enabling NumPy-compatible operations, matrix masking, diagonal reduction, file I/O, and domain-level interaction analysis.

Refer to the sections below for detailed information on how to use BandHiC effectively:

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Web: <https://xdwwb.github.io/BandHiC-Master/>

PyPI: <https://pypi.org/project/bandhic/>

Github: <https://github.com/xdwwb/BandHiC-Master>

QUICK START

This section shows basic usage of the **BandHiC** package, including installation and a brief introduction to its features.

1.1 Overview

BandHiC is a Python package for efficient storage, manipulation, and analysis of Hi-C matrices using a banded matrix representation.

BandHiC adopts a banded storage scheme that stores only a configurable diagonal bandwidth of the dense Hi-C contact matrices. This design can reduce memory usage by up to 99% compared to dense matrices, while still supporting fast random access and user-friendly indexing operations.

Main features include:

- Efficient storage of Hi-C matrices using a banded representation.
- Fast random access and indexing operations.
- Support for flexible masking of missing values, outliers, and unmappable regions.
- Vectorized operations optimized with NumPy for high performance.
- Scalable for ultra-high-resolution Hi-C data analysis.

1.2 Installation

1.2.1 Required Package

BandHiC can be installed on Linux-like systems and requires the following dependencies:

1. python `>= 3.11`
2. numpy `>= 2.3`
3. pandas `>= 2.3`
4. scipy `>= 1.16`
5. cooler `>= 0.10`
6. hic_straw `>= 1.3`

There are two recommended ways to install **BandHiC**:

1.2.2 Option 1: Install via pip

If you already have Python ≥ 3.11 installed:

```
pip install bandhic
```

1.2.3 Option 2: Install from source with conda

```
# 1. Clone the repository
git clone https://github.com/xdwbb/BandHiC-Master.git
cd BandHiC-Master

# 2. Create the environment and activate it
conda env create -f environment.yml
conda activate bandhic

# 3. Install BandHiC
pip install .
```

1.2.4 Build Troubleshooting for hic-straw

If you encounter an error like the following while installing or building hic-straw:

```
fatal error: curl/curl.h: No such file or directory
```

This means the C++ extension in hic-straw requires the **libcurl development headers**, which are not installed by default on many systems.

Solution 1: Install system dependencies (for pip installation)

You need to install the libcurl development package before building:

- On Ubuntu/Debian:

```
sudo apt-get update
sudo apt-get install libcurl4-openssl-dev
```

- On Fedora/CentOS/RHEL:

```
sudo dnf install libcurl-devel
```

- On macOS (with Homebrew):

```
brew install curl
```

If Homebrew's curl is not found automatically, you may need to set environment variables:

```
export CPATH="$(brew --prefix curl)/include"
export LIBRARY_PATH="$(brew --prefix curl)/lib"
```

Solution 2: Use Conda (recommended for convenience)

Instead of building hic-straw from source, you can install a prebuilt binary via [Bioconda](#):

```
conda install -c bioconda hic-straw
```

To avoid conflicts and ensure reproducibility, we recommend installing it in a fresh Conda environment:

```
conda create -n bandhic-env python=3.11
conda activate bandhic-env
conda install -c bioconda hic-straw

# Install BandHiC
pip install bandhic
```

1.3 What is BandHiC?

1.3.1 Data structure of BandHiC

To address the growing memory demands posed by high-resolution Hi-C data, we introduce `band_hic_matrix`, the core class implemented in the BandHiC package. For a Hi-C contact matrix $A \in \mathbb{R}^{n \times n}$ at resolution r , `band_hic_matrix` retains only the diagonals within a user-defined bandwidth k , yielding a compact representation $D \in \mathbb{R}^{n \times k}$. This format ensures that each column in D corresponds to a fixed diagonal of A , such that the mapping $A[i, j] = D[i, j - i]$ holds for $|i - j| \leq k$.

The memory efficiency achieved by this strategy is substantial. When $k \ll n$, the memory footprint of `band_hic_matrix` is reduced from $\mathcal{O}(n^2)$ to $\mathcal{O}(nk)$. For example, assuming a resolution of 1 kb and a bandwidth of 2 Mb ($k = 200$), the representation of chromosome 1 of the human genome (~249 Mb) requires 3.7 GB of memory, less than 1% of the memory required by the dense matrix (~461.9 GB). This compression enables the use of high-resolution Hi-C data on commodity hardware without sacrificing random access efficiency.

To further enhance flexibility of usage, `band_hic_matrix` integrates an optional two-tier masking mechanism. The element-wise mask matrix $M \in \{0, 1\}^{n \times k}$ allows users to selectively ignore missing or outlier contacts, enabling robust statistical estimation on unmasked subsets. Additionally, a bin-level mask $X \in \{0, 1\}^n$ supports the exclusion of entire rows or columns, which is particularly useful for removing repetitive genomic regions devoid of a valid Hi-C signal. These masking features facilitate downstream tasks such as the estimation of average contact intensity at given distances, while confirming statistical validity.

Lastly, a scalar default value d is defined to fill in the undefined entries of A not covered by the band matrix D . This default is typically set to 0, consistent with the assumption that long-range interactions are negligibly sparse. It also ensures that reconstruction of the full matrix A (if required) can be achieved seamlessly by combining D , M , X , and d . Overall, `band_hic_matrix` provides an efficient, flexible data structure for scalable Hi-C data analysis.

1.3.2 Features of BandHiC

A key feature of `band_hic_matrix` is its direct coordinate mapping between the banded matrix B and the full dense matrix A . For any pair of genomic loci (i, j) satisfying the band constraint $|i - j| \leq k$, the interaction frequency $A[i, j]$ can be accessed in constant time via $D[i, j - i]$. This structure ensures random access in $\mathcal{O}(1)$ time, which is critical for performance-sensitive Hi-C analyses, particularly when memory constraints prohibit the use of fully dense matrices. Data access in `band_hic_matrix` is fully consistent with that of a dense matrix, as each entry is accessed via $B[i, j] = D[i, j - i] = A[i, j]$, allowing users to interact with `band_hic_matrix` objects as if they were dense matrices without needing to consider the underlying implementation.

Owing to this random-access capability, `band_hic_matrix` supports NumPy-like indexing semantics, including slicing, boolean indexing, and fancy indexing. This design allows users to easily query local chromatin contacts. For instance, a slice operation such as `B[i:j, i:j]` retrieves a banded submatrix. Combined with the `todense` operation, this enables reconstruction of the dense submatrix for downstream analysis or visualization.

In addition to flexible data access, `band_hic_matrix` also supports a wide range of numerical operations, including element-wise arithmetic operations and reduction operations. These operations are implemented using NumPy for efficiency. Standard reduction operations such as `sum`, `min`, and `max` are supported along conventional axes (rows or columns), as well as along the diagonal axis, which is particularly useful in summarizing interaction intensities by genomic distance. This feature facilitates common Hi-C analyses such as distance-decay profiling.

Taken together, `band_hic_matrix` combines the memory efficiency of a banded storage model with the expressiveness of NumPy’s interface. By mimicking both NumPy’s `ndarray` and `MaskedArray` behaviors, it provides an intuitive and powerful interface for users, substantially lowering the barrier to adoption and promoting integration into existing Hi-C data analysis workflows.

1.3.3 BandHiC reduces the memory consumption of TopDom

To evaluate the effectiveness of BandHiC, we benchmarked the TopDom algorithm on chromosome 1 of mouse embryonic stem cell (mESC) Micro-C data across multiple resolutions using both dense matrix and `band_hic_matrix`. TopDom was reimplemented in Python to support both NumPy’s `ndarray` and BandHiC’s `band_hic_matrix` class. As shown in Fig. 1B, BandHiC substantially reduces memory usage at all tested resolutions. At 1000 bp and 500 bp resolution, the dense matrices require over 72 GB of memory, exceeding the available system memory (60 GB RAM and 12 GB swap space), and causing the program to terminate due to memory allocation failure. In contrast, the banded format completes successfully, with memory usage of only 5,989 MB and 23,902 MB at 1000 bp and 500 bp resolution, respectively. The banded format introduces modest additional runtime compared to the dense matrix (Fig. 1C). This overhead stems not from masking—which was disabled in this evaluation—but from index computation performed during element access in the `band_hic_matrix` object.

These results indicate that `band_hic_matrix` enables domain-calling algorithms like TopDom to be applied to high-resolution Hi-C data on standard hardware, making large-scale analysis feasible without incurring significant computational cost. Owing to BandHiC’s NumPy-like API, other Hi-C-based pattern identification methods can be reimplemented with minimal modifications to the original code, thereby significantly improving their scalability and adaptability to higher-resolution data.

TUTORIALS

This section provides detailed tutorials on how to use the **BandHiC** package effectively.

- *Prerequisites*
- *Import bandhic package*
- *Initialize a band_hic_matrix object*
- *Load or save a band_hic_matrix object*
- *Construct a band_hic_matrix object*
- *Indexing on band_hic_matrix*
- *Masking*
- *Universal functions (ufunc)*
- *Other Array Functions*

2.1 Prerequisites

BandHiC can serve as an alternative to the NumPy package when managing and manipulating Hi-C data, aiming to address the issue of excessive memory usage caused by storing dense matrices using NumPy's `ndarray`. At the same time, BandHiC supports masking operations similar to NumPy's `ma.MaskedArray` module, with enhancements tailored for Hi-C data.

Users can leverage their experience with NumPy when using the BandHiC package, so it is recommended that users have some basic knowledge of NumPy. A link to NumPy is provided below: <https://numpy.org>

2.2 Import bandhic package

```
>>> import bandhic as bh
```

2.3 Initialize a band_hic_matrix object

Initialize from a SciPy `coo_matrix` object:

```
>>> import bandhic as bh
>>> import numpy as np
```

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```
>>> from scipy.sparse import coo_matrix
>>> coo = coo_matrix(([1, 2, 3], ([0, 1, 2],[0, 1, 2])), shape=(3,3))
>>> mat1 = bh.band_hic_matrix(coo, diag_num=2)
```

Initialize from a tuple (data, (row, col)):

```
>>> mat2 = bh.band_hic_matrix(([4, 5, 6], ([0, 1, 2],[2, 1, 0])), diag_num=1)
```

Initialize from a full dense array, only upper-triangular part is stored, lower part is symmetrized:

```
>>> arr = np.arange(16).reshape(4,4)
>>> mat3 = bh.band_hic_matrix(arr, diag_num=3)
```

2.4 Load or save a band_hic_matrix object

```
>>> bh.save_npz('./sample.npz', mat)
>>> mat = bh.load_npz('./sample.npz')
```

Load from .hic file:

```
>>> mat = bh.straw_chr('sample.hic',
                      'chr1',
                      resolution=10000,
                      diag_num=200
                      )
```

Load from .mcool file:

```
>>> mat = bh.cooler_chr('sample.mcool',
                       'chr1',
                       diag_num=200,
                       resolution=10000,
                       )
```

2.5 Construct a band_hic_matrix object

```
# Create a band_hic_matrix object filled with zeros.
>>> mat1 = bh.zeros((5, 5), diag_num=3, dtype=float)

# Create a band_hic_matrix object filled with ones.
>>> mat2 = bh.ones((5, 5), diag_num=3, dtype=float)

# Create a band_hic_matrix object filled as an identity matrix.
>>> mat3 = bh.eye((5, 5), diag_num=3, dtype=float)

# Create a band_hic_matrix object filled with a specified value.
>>> mat4 = bh.full((5, 5), fill_value=0.1, diag_num=3, dtype=float)

# Create a band_hic_matrix object matching another matrix, filled with zeros.
>>> mat5 = bh.zeros_like(mat1, diag_num=3, dtype=float)
```

(continues on next page)


```

# Create a band_hic_matrix object matching another matrix, filled with ones.
>>> mat6 = bh.ones_like(mat1, diag_num=3, dtype=float)

# Create a band_hic_matrix object matching another matrix, filled as an identity matrix.
>>> mat7 = bh.eye_like(mat1, diag_num=3, dtype=float)

# Create a band_hic_matrix object matching another matrix, filled with a specified value.
>>> mat8 = bh.full_like(mat1, fill_value=0.1, diag_num=3, dtype=float)

```

2.6 Indexing on band_hic_matrix

```

# First, we create a band_hic_matrix object:
>>> import numpy as np
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(16).reshape(4,4), diag_num=2)

# Single-element access (scalar)
>>> mat[1, 2]
6

# Masked element returns masked
>>> mat2 = bh.band_hic_matrix(np.eye(4), dtype=int, diag_num=2, mask=([0],[1]))
>>> mat2[0, 1]
masked

# Square submatrix via two-slice indexing returns band_hic_matrix
>>> sub = mat[1:3, 1:3]
>>> isinstance(sub, bh.band_hic_matrix)
True

# Single-axis slice returns band_hic_matrix for square region
>>> sub2 = mat[0:2] # equivalent to mat[0:2, 0:2]
>>> isinstance(sub2, bh.band_hic_matrix)
True

# Fancy indexing returns ndarray or MaskedArray
>>> arr = mat[[0,2,3], [1,2,0]]
>>> isinstance(arr, np.ndarray)
True

>>> mat.add_mask([0,1],[1,2]) # Add mask to some entries
>>> masked_arr = mat[[0,1], [1,2]]
>>> isinstance(masked_arr, np.ma.MaskedArray)
True

# Boolean indexing with band_hic_matrix
>>> mat3 = bh.band_hic_matrix(np.eye(4), diag_num=2, mask=([0,1],[1,2]))
>>> bool_mask = mat3 > 0 # Create a boolean mask
>>> result = mat3[bool_mask] # Use boolean mask for indexing
>>> isinstance(result, np.ma.MaskedArray)

```

(continues on next page)

```
True
>>> result
masked_array(data=[1.0, 1.0, 1.0, 1.0],
             mask=[False, False, False, False],
             fill_value=0.0)
```

2.7 Masking

```
# Add item-wise mask:
>>> mat.add_mask([0, 1], [1, 2])

# Add row/column mask:
>>> mask = np.array([True, False, False])
>>> mat.add_mask_row_col(mask)

# Remove mask for specified indices.
>>> mat.unmask(( [0],[1] ))

# Remove all item-wise mask and row/column mask.
>>> mat.unmask()

# Remove all item-wise mask and row/column mask.
>>> mat.clear_mask()

# Drop all item-wise mask but preserve all row/column mask.
>>> mat.drop_mask()

# Drop all row/column mask.
>>> mat.drop_mask_row_col()

# Access masked `band_hic_matrix` will obtain `np.ma.MaskedArray` object:
>>> mat.add_mask([0, 1], [1, 2])
>>> masked_arr = mat[[0,1], [1,2]]
>>> isinstance(masked_arr, np.ma.MaskedArray)
True
```

2.8 Universal functions (ufunc)

Universal functions that BandHiC supports:

Function 1	Description 1	Function 2	Description 2
absolute	Absolute value	add	Element-wise addition
arccos	Inverse cosine	arccosh	Inverse hyperbolic cosine
arcsin	Inverse sine	arcsinh	Inverse hyperbolic sine
arctan	Inverse tangent	arctan2	Arctangent of y/x with quadrant
arctanh	Inverse hyperbolic tangent	bitwise_and	Element-wise bitwise AND
bitwise_or	Element-wise bitwise OR	bitwise_xor	Element-wise bitwise XOR
cbt	Cube root	conj	Complex conjugate

continues on next page

Table 1 – continued from previous page

Function 1	Description 1	Function 2	Description 2
conjugate	Alias for conj	cos	Cosine function
cosh	Hyperbolic cosine	deg2rad	Degrees to radians
degrees	Radians to degrees	divide	Element-wise division
divmod	Quotient and remainder	equal	Element-wise equality test
exp	Exponential	exp2	Base-2 exponential
expm1	$\exp(x) - 1$	fabs	Absolute value (float)
float_power	Floating-point power	floor_divide	Integer division (floor)
fmod	Modulo operation	gcd	Greatest common divisor
greater	Element-wise greater-than test	greater_equal	Greater-than or equal test
heaviside	Heaviside step function	hypot	Euclidean norm
invert	Bitwise inversion	lcm	Least common multiple
left_shift	Bitwise left shift	less	Element-wise less-than test
less_equal	Less-than or equal test	log	Natural logarithm
log1p	$\log(1 + x)$	log2	Base-2 logarithm
log10	Base-10 logarithm	logaddexp	$\log(\exp(x) + \exp(y))$
logaddexp2	Base-2 version of logaddexp	logical_and	Element-wise logical AND
logical_or	Element-wise logical OR	logical_xor	Element-wise logical XOR
maximum	Element-wise maximum	minimum	Element-wise minimum
mod	Remainder (modulo)	multiply	Element-wise multiplication
negative	Element-wise negation	not_equal	Element-wise inequality test
positive	Returns input unchanged	power	Raise to power
rad2deg	Radians to degrees	radians	Degrees to radians
reciprocal	Element-wise reciprocal	remainder	Modulo remainder
right_shift	Bitwise right shift	rint	Round to nearest integer
sign	Sign of input	sin	Sine function
sinh	Hyperbolic sine	sqrt	Square root
square	Square of input	subtract	Element-wise subtraction
tan	Tangent function	tanh	Hyperbolic tangent
true_divide	Division that returns float		

BandHiC supports these universal functions, and they can be used in the following three ways:

1. As methods of the `band_hic_matrix` object:

```
# When two band_hic_matrix objects are involved, their shape and diag_num must match
>>> mat3 = mat1.add(mat2)
>>> mat4 = mat1.less(mat2)
>>> mat5 = mat1.negative()
```

2. Using mathematical operators:

```
>>> mat3 = mat1 + mat2
>>> mat4 = mat1 < mat2
>>> mat5 = - mat1
```

3. Calling NumPy's universal functions:

```
>>> mat3 = np.add(mat1, mat2)
>>> mat4 = np.less(mat1, mat2)
>>> mat5 = np.negative(mat1)
```

2.9 Other Array Functions

Function	Description
sum	Compute the sum of all elements along the specified axis
prod	Compute the product of all elements along the specified axis
min	Return the minimum value along the specified axis
max	Return the maximum value along the specified axis
mean	Compute the arithmetic mean along the specified axis
var	Compute the variance (average squared deviation)
std	Compute the standard deviation (square root of variance)
ptp	Compute the range (max - min) of values along the axis
all	Return True if all elements evaluate to True
any	Return True if any element evaluates to True
clip	Limit values to a specified min and max range

BandHiC supports these functions, and they can be used in the following two ways:

1. As methods of the `band_hic_matrix` object:

```
# Compute the sum of all elements including out-of-band values filled with `default_
↪value`.
>>> result0 = mat1.sum()

# Compute the sum of all elements along the `row` axis
>>> result1 = mat1.sum(axis=0)
>>> result1 = mat1.sum(axis='row')

# Compute the sum of all elements along the `diag` axis
>>> result2 = mat1.sum(axis='diag')
```

2. Calling NumPy's functions:

```
# Compute the sum of all elements including out-of-band values filled with `default_
↪value`.
>>> result0 = np.sum(mat1)

# Compute the sum of all elements along the `row` axis
>>> result1 = np.sum(mat1, axis=0)

# Compute the sum of all elements along the `diag` axis
>>> result2 = np.sum(mat1, axis='diag')
```

API REFERENCE

This page provides a full API overview of the BandHiC module, including the core `band_hic_matrix` class and its utilities.

The *bandhic* module defines data structures and utilities for efficiently storing and manipulating Hi-C contact matrices in a banded form. It enables NumPy-compatible operations, matrix masking, diagonal reduction, file I/O, and domain-level interaction analysis.

Core components: - *band_hic_matrix*: A memory-efficient matrix class supporting banded Hi-C data. - Utility functions: loaders (*cooler_chr*) and constructors (*ones*, *zeros*, etc.).

3.1 band_hic_matrix Class

The `band_hic_matrix` class is the core data structure in the BandHiC package, designed for efficient storage and manipulation of Hi-C contact matrices using a banded representation. It supports various operations similar to those in NumPy, while optimizing memory usage and performance.

- *Class Overview and Initialization*
- *Masking Methods*
- *Data Indexing and Modification*
- *Data Reduction and Normalization*
- *Vectorized Computation Methods*
- *Data Type Conversion and Representation*
- *Other Methods*

3.1.1 Class Overview and Initialization

class `bandhic.band_hic_matrix`(*contacts*, *diag_num=1*, *mask_row_col=None*, *mask=None*, *dtype=None*, *default_value=0*, *band_data_input=False*)

Symmetric banded matrix stored in upper-triangular format. This storage format is motivated by high-resolution Hi-C data characteristics:

1. Symmetry of contact maps.
2. Interaction frequency concentrated near the diagonal; long-range contacts are sparse (mostly zero).
3. Contact frequency decays sharply with genomic distance.

By storing only the main and a fixed number of super-diagonals as columns of a band matrix (diagonal-major storage: diagonal k stored in column k), we drastically reduce memory usage while enabling random access to Hi-C contacts. Additionally, `mask` and `mask_row_col` arrays track invalid or masked contacts to support downstream analysis.

Operations on this `band_hic_matrix` are as simple as on a `numpy.ndarray`; users can ignore these storage details.

This class stores only the main diagonal and up to $(\text{diag_num} - 1)$ super-diagonals, exploiting symmetry by mirroring values for lower-triangular access.

Parameters

- **contacts** (*coo_array* | *coo_matrix* | *tuple* | *ndarray*)
- **diag_num** (*int*)
- **mask_row_col** (*ndarray* | *None*)
- **mask** (*Tuple[ndarray, ndarray]* | *None*)
- **dtype** (*type* | *None*)
- **default_value** (*int* | *float*)
- **band_data_input** (*bool*)

shape

Shape of the original full Hi-C contact matrix (`bin_num`, `bin_num`), regardless of internal band storage format.

Type

tuple of int

dtype

Data type of the matrix elements, compatible with numpy dtypes.

Type

data-type

diag_num

Number of diagonals stored.

Type

int

bin_num

Number of bins (rows/columns) of the Hi-C matrix.

Type

int

data

Array of shape $(\text{bin_num}, \text{diag_num})$ storing banded Hi-C data.

Type

ndarray

mask

Mask for individual invalid entries. Stored as a boolean ndarray of shape $(\text{bin_num}, \text{diag_num})$ with the same shape as `data`.

Type

ndarray of bool or None

mask_row_col

Mask for entire rows and corresponding columns, indicating invalid bins. Stored as a boolean ndarray of shape (*bin_num*,). For computational convenience, row/column masks are also applied to the *mask* array to track masked entries.

Type

ndarray of bool or None

default_value

Default value for out-of-band entries. Entries out of the banded region and not stored in the data array will be set to this value.

Type

scalar

Examples

```
>>> import bandhic as bh
>>> import numpy as np
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.shape
(4, 4)
```

```
__init__(contacts, diag_num=1, mask_row_col=None, mask=None, dtype=None, default_value=0,
          band_data_input=False)
```

Initialize a `band_hic_matrix` instance.

Parameters

- **contacts** (*{coo_array, coo_matrix, tuple, ndarray}*) – Input Hi-C data in COO format, tuple (data, (row, col)), full square array, or banded stored ndarray. For non-symmetric full arrays, only the upper-triangular part is used and the matrix is symmetrized. Full square arrays are not recommended for large matrices due to memory constraints.
- **diag_num** (*int, optional*) – Number of diagonals to store. Must be ≥ 1 and $\leq$ matrix dimension. Default is 1.
- **mask_row_col** (*ndarray of bool or indices, optional*) – Mask for invalid rows/columns. Can be specified as:
 - A boolean array of shape (*bin_num*,) indicating which rows/columns to mask.
 - A list of indices to mask.Defaults to None (no masking).
- **mask** (*ndarray pair of (row_indices, col_indices), optional*) – Mask for invalid matrix entries. Can be specified as:
 - A tuple of two ndarray (*row_indices*, *col_indices*) listing positions to mask.Defaults to None (no masking).
- **dtype** (*data-type, optional*) – Desired numpy dtype; defaults to ‘contacts’ data dtype; compatible with numpy dtypes.
- **default_value** (*scalar, optional*) – Default value for unstored out-of-band entries. Default is 0.
- **band_data_input** (*bool, optional*) – If True, contacts is treated as precomputed band storage. Default is False.

Raises

ValueError – If contacts type is invalid, diag_num out of range, or array shape invalid.

Return type

None

Examples

Initialize from a SciPy COO matrix:

```
>>> import bandhic as bh
>>> import numpy as np
>>> from scipy.sparse import coo_matrix
>>> coo = coo_matrix(([1, 2, 3], ([0, 1, 2],[0, 1, 2])), shape=(3,3))
>>> mat1 = bh.band_hic_matrix(coo, diag_num=2)
>>> mat1.data.shape
(3, 2)
```

Initialize from a tuple (data, (row, col)):

```
>>> mat2 = bh.band_hic_matrix(([4, 5, 6], ([0, 1, 2],[2, 1, 0])), diag_num=1)
>>> mat2.data.shape
(3, 1)
```

Initialize from a full dense array, only upper-triangular part is stored, lower part is symmetrized:

```
>>> arr = np.arange(16).reshape(4,4)
>>> mat3 = bh.band_hic_matrix(arr, diag_num=3)
>>> mat3.data.shape
(4, 3)
```

Initialize with row/column mask, this masks entire rows and corresponding columns:

```
>>> mask = np.array([True, False, False, True])
>>> mat4 = bh.band_hic_matrix(arr, diag_num=2, mask_row_col=mask)
>>> mat4.mask_row_col
array([ True, False, False,  True])
```

mask_row_col is also supported as a list of indices:

```
>>> mat4 = bh.band_hic_matrix(arr, diag_num=2, mask_row_col=[0, 3])
>>> mat4.mask_row_col
array([ True, False, False,  True])
```

Initialize from precomputed banded storage:

```
>>> band = mat3.data.copy()
>>> mat5 = bh.band_hic_matrix(band, band_data_input=True)
>>> mat5.data.shape
(4, 3)
```

3.1.2 Masking Methods

These methods allow for the application and manipulation of masks within the matrix. Masks can be used to exclude or highlight specific parts of the matrix during computations.

<code>band_hic_matrix.init_mask()</code>	Initialize mask for invalid entries based on matrix shape.
<code>band_hic_matrix.add_mask(row_idx, col_idx)</code>	Add mask entries for specified indices.
<code>band_hic_matrix.get_mask()</code>	Get current mask array.
<code>band_hic_matrix.unmask([indices])</code>	Remove mask entries for specified indices or clear all.
<code>band_hic_matrix.drop_mask()</code>	Clear the current mask by entry-level, but retain the row/column mask.
<code>band_hic_matrix.add_mask_row_col(mask_row_col)</code>	Mask entire rows and corresponding columns.
<code>band_hic_matrix.get_mask_row_col()</code>	Get current row/column mask.
<code>band_hic_matrix.drop_mask_row_col()</code>	Clear the current row/column mask.
<code>band_hic_matrix.clear_mask()</code>	Clear all masks (both entry-level and row/column-level).
<code>band_hic_matrix.count_masked()</code>	Count the number of masked entries in the banded matrix.
<code>band_hic_matrix.count_unmasked()</code>	Count the number of unmasked entries in the banded matrix.
<code>band_hic_matrix.count_in_band_masked()</code>	Count the number of masked entries in the in-band region.
<code>band_hic_matrix.count_out_band_masked()</code>	Count the number of masked entries in the out-of-band region.
<code>band_hic_matrix.count_in_band()</code>	Count the number of valid entries in the in-band region.
<code>band_hic_matrix.count_out_band()</code>	Count the number of valid entries in the out-of-band region.

3.1.2.1 bandhic.band_hic_matrix.init_mask

`band_hic_matrix.init_mask()`

Initialize mask for invalid entries based on matrix shape.

Raises

ValueError – If mask is already initialized.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.init_mask()
```

3.1.2.2 bandhic.band_hic_matrix.add_mask

`band_hic_matrix.add_mask(row_idx, col_idx)`

Add mask entries for specified indices.

Parameters

- **row_idx** (array-like of int) – Row indices to mask.
- **col_idx** (array-like of int) – Column indices to mask.

Raises

ValueError – If row_idx and col_idx have different shapes.

Return type

None

Examples

```
>>> import bandhic as bh
>>> import numpy as np
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.add_mask([0, 1], [1, 2])
```

3.1.2.3 bandhic.band_hic_matrix.get_mask

`band_hic_matrix.get_mask()`

Get current mask array.

Returns

Current mask array or None if no mask.

Return type

ndarray or None

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mask = mat.get_mask()
```

3.1.2.4 bandhic.band_hic_matrix.unmask

`band_hic_matrix.unmask(indices=None)`

Remove mask entries for specified indices or clear all.

Parameters

indices (*tuple of array-like or None*) – Tuple (row_idx, col_idx) to unmask, or None to clear all.

Raises

Warning – If no mask exists when trying to remove.

Return type

None

Notes

If *indices* is None, this will clear all masks (both entry-level and row/column), equivalent to *clear_mask()*.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> mat.unmask(( [0], [0] ))
>>> mat.unmask()
```

3.1.2.5 bandhic.band_hic_matrix.drop_mask

`band_hic_matrix.drop_mask()`

Clear the current mask by entry-level, but retain the row/column mask.

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2, mask=(np.array([[0, 1], [1, 0],
↪ 2]])))
>>> mat.drop_mask()
```

Notes

This method is useful when you want to keep the row/column mask but clear the entry-level mask. It will not affect the row/column mask, allowing you to maintain the masking of entire rows and columns.

3.1.2.6 bandhic.band_hic_matrix.add_mask_row_col

`band_hic_matrix.add_mask_row_col(mask_row_col)`

Mask entire rows and corresponding columns.

Parameters

mask_row_col (*array-like of int or bool*) – If boolean array of shape (bin_num,), True entries indicate rows/columns to mask. If integer array or sequence, treated as indices of rows/columns to mask.

Raises

ValueError – If integer indices are out of bounds.

Return type

None

Examples

```
>>> import bandhic as bh
>>> mask = np.array([True, False, False])
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> mat.add_mask_row_col(mask)
```

3.1.2.7 bandhic.band_hic_matrix.get_mask_row_col

`band_hic_matrix.get_mask_row_col()`

Get current row/column mask.

Returns

Current row/column mask or None if no mask.

Return type

ndarray or None

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mask_row_col = mat.get_mask_row_col()
```

3.1.2.8 bandhic.band_hic_matrix.drop_mask_row_col

`band_hic_matrix.drop_mask_row_col()`

Clear the current row/column mask.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.drop_mask_row_col()
```

3.1.2.9 bandhic.band_hic_matrix.clear_mask

`band_hic_matrix.clear_mask()`

Clear all masks (both entry-level and row/column-level).

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.clear_mask()
```

3.1.2.10 bandhic.band_hic_matrix.count_masked

`band_hic_matrix.count_masked()`

Count the number of masked entries in the banded matrix.

This counts the number of entries in the upper triangular part of the matrix, excluding the diagonal and valid entries.

Returns

Number of valid entries in the banded matrix.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_masked()
0
```

3.1.2.11 bandhic.band_hic_matrix.count_unmasked

`band_hic_matrix.count_unmasked()`

Count the number of unmasked entries in the banded matrix.

Returns

Number of unmasked entries in the banded matrix.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_unmasked()
16
```

3.1.2.12 bandhic.band_hic_matrix.count_in_band_masked

`band_hic_matrix.count_in_band_masked()`

Count the number of masked entries in the in-band region.

Returns

Number of masked entries in the in-band region.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_in_band_masked()
0
```

3.1.2.13 bandhic.band_hic_matrix.count_out_band_masked

`band_hic_matrix.count_out_band_masked()`

Count the number of masked entries in the out-of-band region.

Returns

Number of masked entries in the out-of-band region.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_out_band_masked()
0
```

3.1.2.14 bandhic.band_hic_matrix.count_in_band

`band_hic_matrix.count_in_band()`

Count the number of valid entries in the in-band region.

Returns

Number of valid entries in the in-band region.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_in_band()
10
```

3.1.2.15 bandhic.band_hic_matrix.count_out_band

`band_hic_matrix.count_out_band()`

Count the number of valid entries in the out-of-band region.

Returns

Number of valid entries in the out-of-band region.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2)
>>> mat.count_out_band()
6
```

3.1.3 Data Indexing and Modification

The following methods provide functionality to access, modify, and index the matrix data. These are essential for manipulating individual elements or subsets of the matrix.

<code>band_hic_matrix.__getitem__(index)</code>	Retrieve matrix entries or submatrix using NumPy-like indexing.
<code>band_hic_matrix.__setitem__(index, values)</code>	Assign values to matrix entries using NumPy-like indexing.
<code>band_hic_matrix.get_values(row_idx, col_idx)</code>	Retrieve values considering mask.
<code>band_hic_matrix.set_values(row_idx, col_idx, ...)</code>	Set values at specified row and column indices.
<code>band_hic_matrix.diag(k)</code>	Retrieve the k-th diagonal from the matrix.
<code>band_hic_matrix.set_diag(k, values)</code>	Set values in the k-th diagonal of the matrix.
<code>band_hic_matrix.extract_row(idx[, ...])</code>	Extract stored, unmasked band values for a given row or column.
<code>band_hic_matrix.__iter__()</code>	Iterate over diagonals of the matrix.
<code>band_hic_matrix.iterwindows(width[, step])</code>	Iterate over the diagonals of the matrix with a specified window size.

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Table 2 – continued from previous page

<code>band_hic_matrix.iterrows()</code>	Iterate over the rows of the <code>band_hic_matrix</code> object.
<code>band_hic_matrix.itercols()</code>	Iterate over the columns of the <code>band_hic_matrix</code> object.
<code>band_hic_matrix.filled([fill_value, copy])</code>	Fill masked entries in data with default value.
<code>band_hic_matrix.clip(min_val, max_val)</code>	Clip data values to given range.

3.1.3.1 `bandhic.band_hic_matrix.__getitem__`

`band_hic_matrix.__getitem__(index)`

Retrieve matrix entries or submatrix using NumPy-like indexing.

Supports:

- Integer indexing: `mat[i, j]` returns a single value.
- Slice indexing: `mat[i:j, i:j]` returns a `band_hic_matrix` for square slices.
- Single-axis slice: `mat[i:j]` returns a `band_hic_matrix` same as `mat[i:j, i:j]`.
- Fancy (array) indexing: `mat[[i1, i2], [j1, j2]]` returns an ndarray or MaskedArray according to *mask*.
- Mixed indexing: combinations of integer, slice, and array-like indices.
- Boolean indexing: ‘`band_hic_matrix`’ object with dtype *bool* can be used to index entries.

When both row and column indices specify the same slice (or a single slice is provided), a new `band_hic_matrix` representing that square submatrix is returned. New submatrix is the view of the original matrix, sharing the same data and mask. If the mask or data is altered in the submatrix, the original matrix will reflect those changes as well.

- If a single integer index is provided for both row and column, a scalar value is returned.
- If a mask is set, masked entries will return as *numpy.ma.masked* for scalars, or as a *numpy.ma.MaskedArray* for arrays.
- If a mask is not set, the scalar value is returned directly, or a *numpy.ndarray* for arrays.
- If a square slice is provided, a new `band_hic_matrix` is returned with the same diagonal number and shape as the original matrix.
- If a single slice is provided, it returns a `band_hic_matrix` with the same diagonal number and shape as the original matrix.
- If fancy indexing is used, it returns a *numpy.ndarray* or *numpy.ma.MaskedArray* depending on whether the mask is set.
- In all other cases, a *numpy.ndarray* (if no mask) or *numpy.ma.MaskedArray* (if mask present) is returned.

Parameters

index (*int*, *slice*, *band_hic_matrix*, *array-like of int*, or *tuple of these*)

– index expression for rows and columns. May be:

- A pair (*row_idx*, *col_idx*) of ints, slices, or array-like for mixed indexing.
- A single slice selecting a square region.
- A *band_hic_matrix* object with dtype of *bool* for boolean indexing.

Returns

- scalar : when both row and column are integer indices.
- *numpy.ndarray* : for fancy or mixed indexing without mask.

- `numpy.ma.MaskedArray` : for fancy or mixed indexing when mask is set.
- `band_hic_matrix` : for square slice results.

Return type

scalar or ndarray or MaskedArray or *band_hic_matrix*

Raises

ValueError – If a slice step is not 1, or if indices are out of bounds.

Examples

```
>>> import numpy as np
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(16).reshape(4,4), diag_num=2)
```

Single-element access (scalar)

```
>>> mat[1, 2]
6
```

Masked element returns masked

```
>>> mat2 = bh.band_hic_matrix(np.eye(4), dtype=int, diag_num=2, mask=([0],[1]))
>>> mat2[0, 1]
masked
```

Square submatrix via two-slice indexing returns `band_hic_matrix`

```
>>> sub = mat[1:3, 1:3]
>>> isinstance(sub, bh.band_hic_matrix)
True
```

Single-axis slice returns `band_hic_matrix` for square region

```
>>> sub2 = mat[0:2] # equivalent to mat[0:2, 0:2]
>>> isinstance(sub2, bh.band_hic_matrix)
True
```

Fancy indexing returns ndarray or MaskedArray

```
>>> arr = mat[[0,2,3], [1,2,0]]
>>> isinstance(arr, np.ndarray)
True
```

```
>>> mat.add_mask([0,1],[1,2]) # Add mask to some entries
>>> masked_arr = mat[[0,1], [1,2]]
>>> isinstance(masked_arr, np.ma.MaskedArray)
True
```

Boolean indexing with `band_hic_matrix`

```
>>> mat3 = bh.band_hic_matrix(np.eye(4), diag_num=2, mask=([0,1],[1,2]))
>>> bool_mask = mat3 > 0 # Create a boolean mask
>>> result = mat3[bool_mask] # Use boolean mask for indexing
```

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```
>>> isinstance(result, np.ma.MaskedArray)
True
```

```
>>> result
masked_array(data=[1.0, 1.0, 1.0, 1.0],
              mask=[False, False, False, False],
              fill_value=0.0)
```

3.1.3.2 bandhic.band_hic_matrix.__setitem__

`band_hic_matrix.__setitem__(index, values)`

Assign values to matrix entries using NumPy-like indexing.

Parameters

- **index** (*int, tuple of (row_idx, col_idx), slice, or band_hic_matrix*) – Index expression for rows and columns. May be:
- **column.** (*- A single integer for both row and*)
- **int** (*- A tuple of row and column indices (can be)*)
- **slice**
- **array-like).** (*or*)
- **region.** (*- A single slice selecting a square*)
- **indexing.** (*- A band_hic_matrix object with dtype of bool for boolean*)
- **values** (*scalar or array-like*) – Values to assign. Can be a single scalar or an array-like object.

Raises

- **ValueError** – If index is a slice with step not equal to 1, or if indices exceed matrix dimensions.
- **TypeError** – If *values* is not a scalar or array-like object.
- **Supports:** –
 - - **Integer indexing** – `mat[i, j] = value` assigns to a single element.:
 - - **Slice indexing** – `mat[i:j, i:j] = array or scalar` assigns to a square submatrix.:
 - - **Single-axis slice** – `mat[i:j] = ...` is equivalent to `mat[i:j, i:j].:`
 - - **Fancy (array) indexing** – `mat[[i1, i2], [j1, j2]] = array or scalar` for scattered assignments.:
 - - **Mixed indexing** – combinations of integer, slice, and array-like indices.:
 - - **Boolean indexing** – boolean mask (another `band_hic_matrix` with `dtype=bool`) selects entries to set.:

Return type

None

Examples

```
>>> import bandhic as bh
>>> import numpy as np
>>> mat = bh.band_hic_matrix(np.zeros((4,4)), diag_num=2, dtype=int)
```

Single element assignment

```
>>> mat[1, 2] = 5
>>> mat[1, 2]
5
```

Slice assignment to square submatrix

```
>>> mat[0:2, 0:2] = [[1, 2], [2, 4]]
>>> mat[0:2, 0:2].todense()
array([[1, 2],
       [2, 4]])
```

Single-axis slice assignment (equivalent square slice)

```
>>> mat[2:4] = 0
>>> mat[2:4].todense()
array([[0, 0],
       [0, 0]])
```

Fancy indexing for scattered assignments

```
>>> mat[[0, 3], [1, 2]] = [7, 8]
>>> mat[0, 1], mat[3, 2]
(7, 8)
```

Boolean mask assignment

```
>>> mat2 = bh.band_hic_matrix(np.eye(4), diag_num=2, dtype=int)
>>> bool_mask = mat2 > 0
>>> mat2[bool_mask] = 9
>>> mat2.todense()
array([[9, 0, 0, 0],
       [0, 9, 0, 0],
       [0, 0, 9, 0],
       [0, 0, 0, 9]])
```

Notes

- Assigning to masked entries updates underlying data but does not automatically unmask.
- For multidimensional assignments, scalar values broadcast to all selected positions, while array values must match the number of targeted elements.
- If a boolean mask is used, it must be a *band_hic_matrix* with dtype *bool* and the same shape as the original matrix.
- If a single slice is provided, it behaves like `mat[i:j, i:j]` for square submatrices.

3.1.3.3 bandhic.band_hic_matrix.get_values

`band_hic_matrix.get_values(row_idx, col_idx)`

Retrieve values considering mask.

Parameters

- **row_idx** (*array-like of int*) – Row indices.
- **col_idx** (*array-like of int*) – Column indices.

Returns

Retrieved values; masked entries yield masked results.

Return type

ndarray or MaskedArray

Raises

ValueError – If indices exceed matrix dimensions.

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat.get_values([0,1],[1,2])
array([1., 1.])
```

3.1.3.4 bandhic.band_hic_matrix.set_values

`band_hic_matrix.set_values(row_idx, col_idx, values)`

Set values at specified row and column indices.

Parameters

- **row_idx** (*array-like of int*) – Row indices where values will be set.
- **col_idx** (*array-like of int*) – Column indices where values will be set.
- **values** (*scalar or array-like*) – Values to assign.

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.zeros((3,3), diag_num=2, dtype=int)
>>> mat.set_values([0,1], [1,2], [4,5])
>>> mat[0,1]
4
```

Notes

Writing to masked positions will update the underlying data but will not clear the mask.

3.1.3.5 bandhic.band_hic_matrix.diag

`band_hic_matrix.diag(k)`

Retrieve the *k*-th diagonal from the matrix.

Parameters

k (*int*) – The diagonal index to retrieve.

Returns

The *k*-th diagonal values; masked if mask is set.

Return type

ndarray or MaskedArray

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat.diag(1)
array([1., 1.]
```

3.1.3.6 bandhic.band_hic_matrix.set_diag

`band_hic_matrix.set_diag(k, values)`

Set values in the *k*-th diagonal of the matrix.

Parameters

- ***k*** (*int*) – The diagonal index to set.
- ***values*** (*array-like*) – The values to set in the diagonal.

Raises

ValueError – If *k* is out of range or values length mismatch.

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.zeros((4,4), diag_num=3)
>>> mat.set_diag(1, [9,9,9])
>>> mat.diag(1)
array([9., 9., 9.]
```

3.1.3.7 bandhic.band_hic_matrix.extract_row

`band_hic_matrix.extract_row(idx, extract_out_of_band=True)`

Extract stored, unmasked band values for a given row or column.

Parameters

- ***idx*** (*int*) – Row (or column, due to symmetry) index for which to extract band values.
- ***extract_out_of_band*** (*bool*, *optional*) – If True, include out-of-band entries filled with *default_value* and masked if appropriate. If False (default), return only the stored band values.

Returns

If *extract_out_of_band=False*, a 1D array of length up to *diag_num* containing band values. If *extract_out_of_band=True*, a 1D array of length *bin_num* with all row/column values.

Return type

ndarray or MaskedArray

Raises

ValueError – If *idx* is outside the range [0, *bin_num*-1].

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.extract_row(0, extract_out_of_band=False)
array([0, 1])
>>> mat.extract_row(0)
array([0, 1, 0])
```

3.1.3.8 bandhic.band_hic_matrix.__iter__

`band_hic_matrix.__iter__()`

Iterate over diagonals of the matrix.

Yields

ndarray – Values of each diagonal.

Return type

Iterator[ndarray]

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> for band in mat:
...     print(band)
[1. 1. 1.]
[1. 1.]
```

3.1.3.9 bandhic.band_hic_matrix.iterwindows

`band_hic_matrix.iterwindows(width, step=1)`

Iterate over the diagonals of the matrix with a specified window size.

Parameters

- **width** (*int*) – The size of the window to iterate over.
- **step** (*int*, *optional*) – Step size between windows. Default is 1.

Yields

band_hic_matrix – The values in the current window.

Return type

Iterator[*band_hic_matrix*]

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> for win in mat.iterwindows(2):
...     print(win)
band_hic_matrix(shape=(2, 2), diag_num=2, dtype=<class 'numpy.float64'>)
band_hic_matrix(shape=(2, 2), diag_num=2, dtype=<class 'numpy.float64'>)
```

3.1.3.10 bandhic.band_hic_matrix.iterrows

`band_hic_matrix.iterrows()`

Iterate over the rows of the `band_hic_matrix` object.

Yields

ndarray – The values in the current row.

Return type

Iterator[*ndarray*]

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> for row in mat.iterrows():
...     print(row)
[1. 1. 1.]
[1. 1. 1.]
[1. 1. 1.]
```

3.1.3.11 bandhic.band_hic_matrix.itercols

`band_hic_matrix.itercols()`

Iterate over the columns of the `band_hic_matrix` object.

Yields

ndarray – The values in the current column.

Return type

Iterator[*ndarray*]

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> for col in mat.itercols():
...     print(col)
[1. 1. 1.]
[1. 1. 1.]
[1. 1. 1.]
```

3.1.3.12 bandhic.band_hic_matrix.filled

`band_hic_matrix.filled(fill_value=None, copy=True)`

Fill masked entries in data with default value.

Parameters

- **fill_value** (*scalar*) – Value to assign to masked entries.
- **copy** (*bool*)

Raises

ValueError – If no mask is initialized.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(4), diag_num=2, mask = ([0,1],[1,2]))
>>> mat.filled()
band_hic_matrix(shape=(4, 4), diag_num=2, dtype=float64)
```

3.1.3.13 bandhic.band_hic_matrix.clip

`band_hic_matrix.clip(min_val, max_val)`

Clip data values to given range.

Parameters

- **min_val** (*scalar*)
- **max_val** (*scalar*)

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat = mat.clip(0, 10)
```

3.1.4 Data Reduction and Normalization

These methods enable various data reduction operations, such as summing, averaging, or aggregating matrix values along specific axes or dimensions.

<code>band_hic_matrix.min([axis])</code>	Compute the minimum value in the matrix or along a given axis.
<code>band_hic_matrix.max([axis])</code>	Compute the maximum value in the matrix or along a given axis.
<code>band_hic_matrix.sum([axis])</code>	Compute the sum of the values in the matrix or along a given axis.

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<code>band_hic_matrix.mean([axis])</code>	Compute the mean value of the matrix or along a given axis.
<code>band_hic_matrix.std([axis])</code>	Compute the standard deviation of the values in the matrix or along a given axis.
<code>band_hic_matrix.var([axis])</code>	Compute the variance of the values in the matrix or along a given axis.
<code>band_hic_matrix.prod([axis])</code>	Compute the product of the values in the matrix or along a given axis.
<code>band_hic_matrix.ptp([axis])</code>	Compute the peak-to-peak (maximum - minimum) value of the matrix or along a given axis.
<code>band_hic_matrix.all([axis, banded_only])</code>	Test whether all (or any) array elements along a given axis evaluate to True.
<code>band_hic_matrix.any([axis, banded_only])</code>	Test whether all (or any) array elements along a given axis evaluate to True.
<code>band_hic_matrix.normalize([inplace])</code>	Normalize each diagonal of the matrix to have zero mean and unit variance.

3.1.4.1 bandhic.band_hic_matrix.min

`band_hic_matrix.min(axis=None)`

Compute the minimum value in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row', 'col', 'diag'}, optional*) – Axis along which to compute the minimum:

- None: compute over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Minimum value(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.min() # global minimum
0
>>> mat.min(axis='row')
array([0, 1, 0])
```

3.1.4.2 bandhic.band_hic_matrix.max

`band_hic_matrix.max(axis=None)`

Compute the maximum value in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the maximum:

- None: compute over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Maximum value(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.max() # global maximum
8
>>> mat.max(axis='row')
array([1, 5, 8])
```

3.1.4.3 bandhic.band_hic_matrix.sum

`band_hic_matrix.sum(axis=None)`

Compute the sum of the values in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the sum:

- None: sum over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Sum(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.sum() # sum of all elements
24
>>> mat.sum(axis='row')
array([ 1, 10, 13])
```

3.1.4.4 bandhic.band_hic_matrix.mean

`band_hic_matrix.mean(axis=None)`

Compute the mean value of the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the mean:

- None: mean over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Mean value(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.mean() # mean of all elements
2.6666666666666665
>>> mat.mean(axis='diag')
array([4., 3.])
```

3.1.4.5 bandhic.band_hic_matrix.std

`band_hic_matrix.std(axis=None)`

Compute the standard deviation of the values in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the standard deviation:

- None: std over all stored values (and default for missing).

- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Standard deviation(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.std() # std of all elements
2.748737083745107
>>> mat.std(axis='diag')
array([3.26598632, 2.          ])
```

3.1.4.6 bandhic.band_hic_matrix.var

`band_hic_matrix.var(axis=None)`

Compute the variance of the values in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row', 'col', 'diag'}, optional*) – Axis along which to compute the variance:

- None: variance over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Variance(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.var() # variance of all elements
7.555555555555555
```

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```
>>> mat.var(axis='row')
array([ 0.22222222,  2.88888889, 10.88888889])
```

3.1.4.7 bandhic.band_hic_matrix.prod

`band_hic_matrix.prod(axis=None)`

Compute the product of the values in the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the product:

- None: product over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Product(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(1, 10).reshape(3,3), diag_num=2)
>>> mat.prod() # product of all elements
0
>>> mat.prod(axis='row')
array([ 0, 60, 0])
```

3.1.4.8 bandhic.band_hic_matrix.ptp

`band_hic_matrix.ptp(axis=None)`

Compute the peak-to-peak (maximum - minimum) value of the matrix or along a given axis.

Parameters

axis (*None, int, or {'row','col','diag'}, optional*) – Axis along which to compute the peak-to-peak value:

- None: ptp over all stored values (and default for missing).
- 0 or 'row': per-row reduction.
- 1 or 'col': per-column reduction.
- 'diag': per-diagonal reduction.

Default is None.

Returns

Peak-to-peak value(s) along the specified axis.

Return type

scalar or ndarray

Raises

ValueError – If axis is not one of the supported values.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat.ptp() # ptp of all elements
8
>>> mat.ptp(axis='row')
array([1, 4, 8])
```

3.1.4.9 bandhic.band_hic_matrix.all

`band_hic_matrix.all(axis=None, banded_only=False)`

Test whether all (or any) array elements along a given axis evaluate to True.

Parameters

- **axis** (*None, int, or {'row','col','diag'}, optional*) – Axis along which to test.
 - None: test all stored values (and default for missing).
 - 0 or 'row': per-row reduction.
 - 1 or 'col': per-column reduction.
 - 'diag': per-diagonal reduction.
- **banded_only** (*bool, optional*) – If True, only consider stored band elements; ignore out-of-band values. Default is False.

Returns

Boolean result(s) of the test.

Return type

bool or ndarray

Raises

ValueError – If axis is not supported.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> mat.all()
False
>>> mat.any(axis='diag', banded_only=True)
array([ True, False])
```

3.1.4.10 bandhic.band_hic_matrix.any

`band_hic_matrix.any(axis=None, banded_only=False)`

Test whether all (or any) array elements along a given axis evaluate to True.

Parameters

- **axis** (*None, int, or {'row','col','diag'}, optional*) – Axis along which to test:
 - None: test all stored values (and default for missing).
 - 0 or 'row': per-row reduction.
 - 1 or 'col': per-column reduction.
 - 'diag': per-diagonal reduction.
- **banded_only** (*bool, optional*) – If True, only consider stored band elements; ignore out-of-band values. Default is False.

Returns

Boolean result(s) of the test.

Return type

bool or ndarray

Raises

ValueError – If axis is not supported.

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> mat.all()
False
>>> mat.any(axis='diag', banded_only=True)
array([ True, False])
```

3.1.4.11 bandhic.band_hic_matrix.normalize

`band_hic_matrix.normalize(inplace=False)`

Normalize each diagonal of the matrix to have zero mean and unit variance. This modifies the matrix in place.
:raises UserWarning: If any diagonal has zero standard deviation, it will be set to zero.

Return type

None

Parameters

inplace (*bool*)

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.arange(9).reshape(3,3), diag_num=2)
>>> mat = mat.normalize()
```

3.1.5 Vectorized Computation Methods

These methods allow for the application of universal functions (ufuncs) to the matrix, enabling element-wise operations similar to those in NumPy.

<code>band_hic_matrix.absolute(self, *args, **kwargs)</code>	Perform element-wise 'absolute' operation.
<code>band_hic_matrix.add(self, other, *args, **kwargs)</code>	Perform element-wise 'add' operation with two inputs.
<code>band_hic_matrix.arccos(self, *args, **kwargs)</code>	Perform element-wise 'arccos' operation.
<code>band_hic_matrix.arccosh(self, *args, **kwargs)</code>	Perform element-wise 'arccosh' operation.
<code>band_hic_matrix.arcsin(self, *args, **kwargs)</code>	Perform element-wise 'arcsin' operation.
<code>band_hic_matrix.arcsinh(self, *args, **kwargs)</code>	Perform element-wise 'arcsinh' operation.
<code>band_hic_matrix.arctan(self, *args, **kwargs)</code>	Perform element-wise 'arctan' operation.
<code>band_hic_matrix.arctan2(self, other, *args, ...)</code>	Perform element-wise 'arctan2' operation with two inputs.
<code>band_hic_matrix.arctanh(self, *args, **kwargs)</code>	Perform element-wise 'arctanh' operation.
<code>band_hic_matrix.bitwise_and(self, other, ...)</code>	Perform element-wise 'bitwise_and' operation with two inputs.
<code>band_hic_matrix.bitwise_or(self, other, ...)</code>	Perform element-wise 'bitwise_or' operation with two inputs.
<code>band_hic_matrix.bitwise_xor(self, other, ...)</code>	Perform element-wise 'bitwise_xor' operation with two inputs.
<code>band_hic_matrix.cbrt(self, *args, **kwargs)</code>	Perform element-wise 'cbrt' operation.
<code>band_hic_matrix.conj(self, *args, **kwargs)</code>	Perform element-wise 'conj' operation.
<code>band_hic_matrix.conjugate(self, *args, **kwargs)</code>	Perform element-wise 'conjugate' operation.
<code>band_hic_matrix.cos(self, *args, **kwargs)</code>	Perform element-wise 'cos' operation.
<code>band_hic_matrix.cosh(self, *args, **kwargs)</code>	Perform element-wise 'cosh' operation.
<code>band_hic_matrix.deg2rad(self, *args, **kwargs)</code>	Perform element-wise 'deg2rad' operation.
<code>band_hic_matrix.degrees(self, *args, **kwargs)</code>	Perform element-wise 'degrees' operation.
<code>band_hic_matrix.divide(self, other, *args, ...)</code>	Perform element-wise 'divide' operation with two inputs.
<code>band_hic_matrix.divmod(self, other, *args, ...)</code>	Perform element-wise 'divmod' operation with two inputs.
<code>band_hic_matrix.equal(self, other, *args, ...)</code>	Perform element-wise 'equal' operation with two inputs.
<code>band_hic_matrix.exp(self, *args, **kwargs)</code>	Perform element-wise 'exp' operation.
<code>band_hic_matrix.exp2(self, *args, **kwargs)</code>	Perform element-wise 'exp2' operation.
<code>band_hic_matrix.expml(self, *args, **kwargs)</code>	Perform element-wise 'expml' operation.
<code>band_hic_matrix.fabs(self, *args, **kwargs)</code>	Perform element-wise 'fabs' operation.
<code>band_hic_matrix.float_power(self, other, ...)</code>	Perform element-wise 'float_power' operation with two inputs.
<code>band_hic_matrix.floor_divide(self, other, ...)</code>	Perform element-wise 'floor_divide' operation with two inputs.
<code>band_hic_matrix.fmod(self, other, *args, ...)</code>	Perform element-wise 'fmod' operation with two inputs.
<code>band_hic_matrix.gcd(self, other, *args, **kwargs)</code>	Perform element-wise 'gcd' operation with two inputs.
<code>band_hic_matrix.greater(self, other, *args, ...)</code>	Perform element-wise 'greater' operation with two inputs.
<code>band_hic_matrix.greater_equal(self, other, ...)</code>	Perform element-wise 'greater_equal' operation with two inputs.
<code>band_hic_matrix.heaviside(self, other, ...)</code>	Perform element-wise 'heaviside' operation with two inputs.
<code>band_hic_matrix.hypot(self, other, *args, ...)</code>	Perform element-wise 'hypot' operation with two inputs.
<code>band_hic_matrix.invert(self, *args, **kwargs)</code>	Perform element-wise 'invert' operation.
<code>band_hic_matrix.lcm(self, other, *args, **kwargs)</code>	Perform element-wise 'lcm' operation with two inputs.
<code>band_hic_matrix.left_shift(self, other, ...)</code>	Perform element-wise 'left_shift' operation with two inputs.

continues on next page

Table 4 – continued from previous page

<code>band_hic_matrix.less(self, other, *args, ...)</code>	Perform element-wise 'less' operation with two inputs.
<code>band_hic_matrix.less_equal(self, other, ...)</code>	Perform element-wise 'less_equal' operation with two inputs.
<code>band_hic_matrix.log(self, *args, **kwargs)</code>	Perform element-wise 'log' operation.
<code>band_hic_matrix.log1p(self, *args, **kwargs)</code>	Perform element-wise 'log1p' operation.
<code>band_hic_matrix.log2(self, *args, **kwargs)</code>	Perform element-wise 'log2' operation.
<code>band_hic_matrix.log10(self, *args, **kwargs)</code>	Perform element-wise 'log10' operation.
<code>band_hic_matrix.logaddexp(self, other, ...)</code>	Perform element-wise 'logaddexp' operation with two inputs.
<code>band_hic_matrix.logaddexp2(self, other, ...)</code>	Perform element-wise 'logaddexp2' operation with two inputs.
<code>band_hic_matrix.logical_and(self, other, ...)</code>	Perform element-wise 'logical_and' operation with two inputs.
<code>band_hic_matrix.logical_or(self, other, ...)</code>	Perform element-wise 'logical_or' operation with two inputs.
<code>band_hic_matrix.logical_xor(self, other, ...)</code>	Perform element-wise 'logical_xor' operation with two inputs.
<code>band_hic_matrix.maximum(self, other, *args, ...)</code>	Perform element-wise 'maximum' operation with two inputs.
<code>band_hic_matrix.minimum(self, other, *args, ...)</code>	Perform element-wise 'minimum' operation with two inputs.
<code>band_hic_matrix.mod(self, other, *args, **kwargs)</code>	Perform element-wise 'mod' operation with two inputs.
<code>band_hic_matrix.multiply(self, other, *args, ...)</code>	Perform element-wise 'multiply' operation with two inputs.
<code>band_hic_matrix.negative(self, *args, **kwargs)</code>	Perform element-wise 'negative' operation.
<code>band_hic_matrix.not_equal(self, other, ...)</code>	Perform element-wise 'not_equal' operation with two inputs.
<code>band_hic_matrix.positive(self, *args, **kwargs)</code>	Perform element-wise 'positive' operation.
<code>band_hic_matrix.power(self, other, *args, ...)</code>	Perform element-wise 'power' operation with two inputs.
<code>band_hic_matrix.rad2deg(self, *args, **kwargs)</code>	Perform element-wise 'rad2deg' operation.
<code>band_hic_matrix.radians(self, *args, **kwargs)</code>	Perform element-wise 'radians' operation.
<code>band_hic_matrix.reciprocal(self, *args, **kwargs)</code>	Perform element-wise 'reciprocal' operation.
<code>band_hic_matrix.remainder(self, other, ...)</code>	Perform element-wise 'remainder' operation with two inputs.
<code>band_hic_matrix.right_shift(self, other, ...)</code>	Perform element-wise 'right_shift' operation with two inputs.
<code>band_hic_matrix rint(self, *args, **kwargs)</code>	Perform element-wise 'rint' operation.
<code>band_hic_matrix.sign(self, *args, **kwargs)</code>	Perform element-wise 'sign' operation.
<code>band_hic_matrix.sin(self, *args, **kwargs)</code>	Perform element-wise 'sin' operation.
<code>band_hic_matrix.sinh(self, *args, **kwargs)</code>	Perform element-wise 'sinh' operation.
<code>band_hic_matrix.sqrt(self, *args, **kwargs)</code>	Perform element-wise 'sqrt' operation.
<code>band_hic_matrix.square(self, *args, **kwargs)</code>	Perform element-wise 'square' operation.
<code>band_hic_matrix.subtract(self, other, *args, ...)</code>	Perform element-wise 'subtract' operation with two inputs.
<code>band_hic_matrix.tan(self, *args, **kwargs)</code>	Perform element-wise 'tan' operation.
<code>band_hic_matrix.tanh(self, *args, **kwargs)</code>	Perform element-wise 'tanh' operation.
<code>band_hic_matrix.true_divide(self, other, ...)</code>	Perform element-wise 'true_divide' operation with two inputs.

3.1.5.1 bandhic.band_hic_matrix.absolute

`band_hic_matrix.absolute(self, *args, **kwargs)`

Perform element-wise ‘absolute’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.absolute`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.absolute`.

Returns

Result of element-wise ‘absolute’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.absolute`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.absolute()
```

3.1.5.2 bandhic.band_hic_matrix.add

`band_hic_matrix.add(self, other, *args, **kwargs)`

Perform element-wise ‘add’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.add`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.add`.

Returns

Result of element-wise ‘add’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.add`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.add(other)
```

3.1.5.3 bandhic.band_hic_matrix.arccos

`band_hic_matrix.arccos(self, *args, **kwargs)`

Perform element-wise ‘arccos’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arccos`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arccos`.

Returns

Result of element-wise ‘arccos’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arccos`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arccos()
```

3.1.5.4 bandhic.band_hic_matrix.arccosh

`band_hic_matrix.arccosh(self, *args, **kwargs)`

Perform element-wise ‘arccosh’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arccosh`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arccosh`.

Returns

Result of element-wise ‘arccosh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arccosh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arccosh()
```

3.1.5.5 bandhic.band_hic_matrix.arcsin

`band_hic_matrix.arcsin(self, *args, **kwargs)`

Perform element-wise ‘arcsin’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arcsin`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arcsin`.

Returns

Result of element-wise ‘arcsin’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arcsin`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arcsin()
```

3.1.5.6 bandhic.band_hic_matrix.arcsinh

`band_hic_matrix.arcsinh(self, *args, **kwargs)`

Perform element-wise ‘arcsinh’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arcsinh`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arcsinh`.

Returns

Result of element-wise ‘arcsinh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arcsinh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arcsinh()
```

3.1.5.7 bandhic.band_hic_matrix.arctan

`band_hic_matrix.arctan(self, *args, **kwargs)`

Perform element-wise ‘arctan’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arctan`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arctan`.

Returns

Result of element-wise ‘arctan’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arctan`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arctan()
```

3.1.5.8 bandhic.band_hic_matrix.arctan2

`band_hic_matrix.arctan2(self, other, *args, **kwargs)`

Perform element-wise ‘arctan2’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arctan2`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arctan2`.

Returns

Result of element-wise ‘arctan2’ operation.

Return type

band_hic_matrix

 See also

`numpy.arctan2`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.arctan2(other)
```

3.1.5.9 `bandhic.band_hic_matrix.arctanh`

`band_hic_matrix.arctanh(self, *args, **kwargs)`

Perform element-wise ‘arctanh’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arctanh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arctanh`.

Returns

Result of element-wise ‘arctanh’ operation.

Return type

band_hic_matrix

 See also

`numpy.arctanh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.arctanh()
```

3.1.5.10 `bandhic.band_hic_matrix.bitwise_and`

`band_hic_matrix.bitwise_and(self, other, *args, **kwargs)`

Perform element-wise ‘bitwise_and’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.bitwise_and`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.bitwise_and`.

Returns

Result of element-wise ‘bitwise_and’ operation.

Return type

band_hic_matrix

 See also

`numpy.bitwise_and`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.bitwise_and(other)
```

3.1.5.11 bandhic.band_hic_matrix.bitwise_or

`band_hic_matrix.bitwise_or(self, other, *args, **kwargs)`

Perform element-wise ‘bitwise_or’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.bitwise_or`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.bitwise_or`.

Returns

Result of element-wise ‘bitwise_or’ operation.

Return type

band_hic_matrix

 See also

`numpy.bitwise_or`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.bitwise_or(other)
```

3.1.5.12 bandhic.band_hic_matrix.bitwise_xor

`band_hic_matrix.bitwise_xor(self, other, *args, **kwargs)`

Perform element-wise ‘bitwise_xor’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.

- ***args** (*tuple*) – Additional positional arguments for `numpy.bitwise_xor`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.bitwise_xor`.

Returns

Result of element-wise ‘bitwise_xor’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.bitwise_xor`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.bitwise_xor(other)
```

3.1.5.13 `bandhic.band_hic_matrix.cbrt`

`band_hic_matrix.cbrt(self, *args, **kwargs)`

Perform element-wise ‘cbrt’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cbrt`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cbrt`.

Returns

Result of element-wise ‘cbrt’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.cbrt`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.cbrt()
```

3.1.5.14 `bandhic.band_hic_matrix.conj`

`band_hic_matrix.conj(self, *args, **kwargs)`

Perform element-wise ‘conj’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.conj`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.conj`.

Returns

Result of element-wise ‘conj’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.conj`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.conj()
```

3.1.5.15 `bandhic.band_hic_matrix.conjugate`

`band_hic_matrix.conjugate(self, *args, **kwargs)`

Perform element-wise ‘conjugate’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.conjugate`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.conjugate`.

Returns

Result of element-wise ‘conjugate’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.conjugate`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.conjugate()
```

3.1.5.16 `bandhic.band_hic_matrix.cos`

`band_hic_matrix.cos(self, *args, **kwargs)`

Perform element-wise ‘cos’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cos`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cos`.

Returns

Result of element-wise ‘cos’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.cos`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.cos()
```

3.1.5.17 `bandhic.band_hic_matrix.cosh`

`band_hic_matrix.cosh(self, *args, **kwargs)`

Perform element-wise ‘cosh’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cosh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cosh`.

Returns

Result of element-wise ‘cosh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.cosh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.cosh()
```

3.1.5.18 bandhic.band_hic_matrix.deg2rad

`band_hic_matrix.deg2rad(self, *args, **kwargs)`

Perform element-wise ‘deg2rad’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.deg2rad`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.deg2rad`.

Returns

Result of element-wise ‘deg2rad’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.deg2rad`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.deg2rad()
```

3.1.5.19 bandhic.band_hic_matrix.degrees

`band_hic_matrix.degrees(self, *args, **kwargs)`

Perform element-wise ‘degrees’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.degrees`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.degrees`.

Returns

Result of element-wise ‘degrees’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.degrees`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.degrees()
```

3.1.5.20 `bandhic.band_hic_matrix.divide`

`band_hic_matrix.divide(self, other, *args, **kwargs)`

Perform element-wise ‘divide’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.divide`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.divide`.

Returns

Result of element-wise ‘divide’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.divide`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.divide(other)
```

3.1.5.21 `bandhic.band_hic_matrix.divmod`

`band_hic_matrix.divmod(self, other, *args, **kwargs)`

Perform element-wise ‘divmod’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.divmod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.divmod`.

Returns

Result of element-wise ‘divmod’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.divmod`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.divmod(other)
```

3.1.5.22 bandhic.band_hic_matrix.equal

`band_hic_matrix.equal(self, other, *args, **kwargs)`

Perform element-wise ‘equal’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.equal`.

Returns

Result of element-wise ‘equal’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.equal`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.equal(other)
```

3.1.5.23 bandhic.band_hic_matrix.exp

`band_hic_matrix.exp(self, *args, **kwargs)`

Perform element-wise ‘exp’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.exp`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.exp`.

Returns

Result of element-wise ‘exp’ operation.

Return type

band_hic_matrix

See also

[numpy.exp](#)

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.exp()
```

3.1.5.24 bandhic.band_hic_matrix.exp2

`band_hic_matrix.exp2(self, *args, **kwargs)`

Perform element-wise ‘exp2’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.exp2`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.exp2`.

Returns

Result of element-wise ‘exp2’ operation.

Return type

band_hic_matrix

See also

[numpy.exp2](#)

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.exp2()
```

3.1.5.25 bandhic.band_hic_matrix.expm1

`band_hic_matrix.expm1(self, *args, **kwargs)`

Perform element-wise ‘expm1’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.expm1`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.expm1`.

Returns

Result of element-wise ‘expm1’ operation.

Return type

band_hic_matrix

 See also

`numpy.expm1`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.expm1()
```

3.1.5.26 `bandhic.band_hic_matrix.fabs`

`band_hic_matrix.fabs(self, *args, **kwargs)`

Perform element-wise ‘fabs’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.fabs`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.fabs`.

Returns

Result of element-wise ‘fabs’ operation.

Return type

band_hic_matrix

 See also

`numpy.fabs`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.fabs()
```

3.1.5.27 `bandhic.band_hic_matrix.float_power`

`band_hic_matrix.float_power(self, other, *args, **kwargs)`

Perform element-wise ‘float_power’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.float_power`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.float_power`.

Returns

Result of element-wise ‘float_power’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.float_power`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.float_power(other)
```

3.1.5.28 `bandhic.band_hic_matrix.floor_divide`

`band_hic_matrix.floor_divide(self, other, *args, **kwargs)`

Perform element-wise ‘floor_divide’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.floor_divide`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.floor_divide`.

Returns

Result of element-wise ‘floor_divide’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.floor_divide`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.floor_divide(other)
```

3.1.5.29 `bandhic.band_hic_matrix.fmod`

`band_hic_matrix.fmod(self, other, *args, **kwargs)`

Perform element-wise ‘fmod’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.

- ***args** (*tuple*) – Additional positional arguments for `numpy.fmod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.fmod`.

Returns

Result of element-wise ‘fmod’ operation.

Return type

band_hic_matrix

 See also

`numpy.fmod`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.fmod(other)
```

3.1.5.30 `bandhic.band_hic_matrix.gcd`

`band_hic_matrix.gcd(self, other, *args, **kwargs)`

Perform element-wise ‘gcd’ operation with two inputs.

Parameters

- **self** (*band_hic_matrix*) – First input matrix.
- **other** (*band_hic_matrix* or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.gcd`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.gcd`.

Returns

Result of element-wise ‘gcd’ operation.

Return type

band_hic_matrix

 See also

`numpy.gcd`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.gcd(other)
```

3.1.5.31 `bandhic.band_hic_matrix.greater`

`band_hic_matrix.greater(self, other, *args, **kwargs)`

Perform element-wise ‘greater’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.greater`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.greater`.

Returns

Result of element-wise ‘greater’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.greater`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.greater(other)
```

3.1.5.32 `bandhic.band_hic_matrix.greater_equal`

`band_hic_matrix.greater_equal(self, other, *args, **kwargs)`

Perform element-wise ‘greater_equal’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.greater_equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.greater_equal`.

Returns

Result of element-wise ‘greater_equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.greater_equal`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.greater_equal(other)
```

3.1.5.33 bandhic.band_hic_matrix.heaviside

`band_hic_matrix.heaviside(self, other, *args, **kwargs)`

Perform element-wise ‘heaviside’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.heaviside`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.heaviside`.

Returns

Result of element-wise ‘heaviside’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.heaviside`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.heaviside(other)
```

3.1.5.34 bandhic.band_hic_matrix.hypot

`band_hic_matrix.hypot(self, other, *args, **kwargs)`

Perform element-wise ‘hypot’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.hypot`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.hypot`.

Returns

Result of element-wise ‘hypot’ operation.

Return type

band_hic_matrix

See also

[numpy.hypot](#)

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.hypot(other)
```

3.1.5.35 `bandhic.band_hic_matrix.invert`

`band_hic_matrix.invert(self, *args, **kwargs)`

Perform element-wise ‘invert’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.invert`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.invert`.

Returns

Result of element-wise ‘invert’ operation.

Return type

band_hic_matrix

See also

[numpy.invert](#)

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.invert()
```

3.1.5.36 `bandhic.band_hic_matrix.lcm`

`band_hic_matrix.lcm(self, other, *args, **kwargs)`

Perform element-wise ‘lcm’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.lcm`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.lcm`.

Returns

Result of element-wise ‘lcm’ operation.

Return type

band_hic_matrix

 See also

`numpy.lcm`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.lcm(other)
```

3.1.5.37 bandhic.band_hic_matrix.left_shift

`band_hic_matrix.left_shift(self, other, *args, **kwargs)`

Perform element-wise ‘left_shift’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.left_shift`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.left_shift`.

Returns

Result of element-wise ‘left_shift’ operation.

Return type

band_hic_matrix

 See also

`numpy.left_shift`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.left_shift(other)
```

3.1.5.38 bandhic.band_hic_matrix.less

`band_hic_matrix.less(self, other, *args, **kwargs)`

Perform element-wise ‘less’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.

- ***args** (*tuple*) – Additional positional arguments for `numpy.less`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.less`.

Returns

Result of element-wise ‘less’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.less`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.less(other)
```

3.1.5.39 `bandhic.band_hic_matrix.less_equal`

`band_hic_matrix.less_equal(self, other, *args, **kwargs)`

Perform element-wise ‘less_equal’ operation with two inputs.

Parameters

- **self** (*band_hic_matrix*) – First input matrix.
- **other** (*band_hic_matrix* or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.less_equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.less_equal`.

Returns

Result of element-wise ‘less_equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.less_equal`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.less_equal(other)
```

3.1.5.40 bandhic.band_hic_matrix.log

`band_hic_matrix.log(self, *args, **kwargs)`

Perform element-wise ‘log’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log`.

Returns

Result of element-wise ‘log’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.log()
```

3.1.5.41 bandhic.band_hic_matrix.log1p

`band_hic_matrix.log1p(self, *args, **kwargs)`

Perform element-wise ‘log1p’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log1p`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log1p`.

Returns

Result of element-wise ‘log1p’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log1p`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.log1p()
```

3.1.5.42 bandhic.band_hic_matrix.log2

`band_hic_matrix.log2(self, *args, **kwargs)`

Perform element-wise ‘log2’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log2`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log2`.

Returns

Result of element-wise ‘log2’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log2`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.log2()
```

3.1.5.43 bandhic.band_hic_matrix.log10

`band_hic_matrix.log10(self, *args, **kwargs)`

Perform element-wise ‘log10’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log10`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log10`.

Returns

Result of element-wise ‘log10’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log10`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.log10()
```

3.1.5.44 `bandhic.band_hic_matrix.logaddexp`

`band_hic_matrix.logaddexp(self, other, *args, **kwargs)`

Perform element-wise ‘logaddexp’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logaddexp`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logaddexp`.

Returns

Result of element-wise ‘logaddexp’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logaddexp`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.logaddexp(other)
```

3.1.5.45 `bandhic.band_hic_matrix.logaddexp2`

`band_hic_matrix.logaddexp2(self, other, *args, **kwargs)`

Perform element-wise ‘logaddexp2’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logaddexp2`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logaddexp2`.

Returns

Result of element-wise ‘logaddexp2’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logaddexp2`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.logaddexp2(other)
```

3.1.5.46 bandhic.band_hic_matrix.logical_and

`band_hic_matrix.logical_and(self, other, *args, **kwargs)`

Perform element-wise ‘logical_and’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_and`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_and`.

Returns

Result of element-wise ‘logical_and’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.logical_and`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.logical_and(other)
```

3.1.5.47 bandhic.band_hic_matrix.logical_or

`band_hic_matrix.logical_or(self, other, *args, **kwargs)`

Perform element-wise ‘logical_or’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_or`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_or`.

Returns

Result of element-wise ‘logical_or’ operation.

Return type

band_hic_matrix

 See also

`numpy.logical_or`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.logical_or(other)
```

3.1.5.48 bandhic.band_hic_matrix.logical_xor

`band_hic_matrix.logical_xor(self, other, *args, **kwargs)`

Perform element-wise ‘logical_xor’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_xor`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_xor`.

Returns

Result of element-wise ‘logical_xor’ operation.

Return type

band_hic_matrix

 See also

`numpy.logical_xor`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.logical_xor(other)
```

3.1.5.49 bandhic.band_hic_matrix.maximum

`band_hic_matrix.maximum(self, other, *args, **kwargs)`

Perform element-wise ‘maximum’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.maximum`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.maximum`.

Returns

Result of element-wise ‘maximum’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.maximum`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.maximum(other)
```

3.1.5.50 `bandhic.band_hic_matrix.minimum`

`band_hic_matrix.minimum(self, other, *args, **kwargs)`

Perform element-wise ‘minimum’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.minimum`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.minimum`.

Returns

Result of element-wise ‘minimum’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.minimum`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.minimum(other)
```

3.1.5.51 `bandhic.band_hic_matrix.mod`

`band_hic_matrix.mod(self, other, *args, **kwargs)`

Perform element-wise ‘mod’ operation with two inputs.

Parameters

- **self** (*band_hic_matrix*) – First input matrix.
- **other** (*band_hic_matrix* or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.mod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.mod`.

Returns

Result of element-wise ‘mod’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.mod`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.mod(other)
```

3.1.5.52 `bandhic.band_hic_matrix.multiply`

`band_hic_matrix.multiply(self, other, *args, **kwargs)`

Perform element-wise ‘multiply’ operation with two inputs.

Parameters

- **self** (*band_hic_matrix*) – First input matrix.
- **other** (*band_hic_matrix* or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.multiply`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.multiply`.

Returns

Result of element-wise ‘multiply’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.multiply`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.multiply(other)
```

3.1.5.53 bandhic.band_hic_matrix.negative

`band_hic_matrix.negative(self, *args, **kwargs)`

Perform element-wise ‘negative’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.negative`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.negative`.

Returns

Result of element-wise ‘negative’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.negative`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.negative()
```

3.1.5.54 bandhic.band_hic_matrix.not_equal

`band_hic_matrix.not_equal(self, other, *args, **kwargs)`

Perform element-wise ‘not_equal’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.not_equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.not_equal`.

Returns

Result of element-wise ‘not_equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.not_equal`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.not_equal(other)
```

3.1.5.55 bandhic.band_hic_matrix.positive

`band_hic_matrix.positive(self, *args, **kwargs)`

Perform element-wise ‘positive’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.positive`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.positive`.

Returns

Result of element-wise ‘positive’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.positive`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.positive()
```

3.1.5.56 bandhic.band_hic_matrix.power

`band_hic_matrix.power(self, other, *args, **kwargs)`

Perform element-wise ‘power’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.power`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.power`.

Returns

Result of element-wise ‘power’ operation.

Return type

band_hic_matrix

See also

`numpy.power`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.power(other)
```

3.1.5.57 `bandhic.band_hic_matrix.rad2deg`

`band_hic_matrix.rad2deg(self, *args, **kwargs)`

Perform element-wise ‘rad2deg’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.rad2deg`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.rad2deg`.

Returns

Result of element-wise ‘rad2deg’ operation.

Return type

band_hic_matrix

See also

`numpy.rad2deg`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.rad2deg()
```

3.1.5.58 `bandhic.band_hic_matrix.radians`

`band_hic_matrix.radians(self, *args, **kwargs)`

Perform element-wise ‘radians’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.radians`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.radians`.

Returns

Result of element-wise ‘radians’ operation.

Return type

band_hic_matrix

 See also

`numpy.radians`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.radians()
```

3.1.5.59 bandhic.band_hic_matrix.reciprocal

`band_hic_matrix.reciprocal(self, *args, **kwargs)`

Perform element-wise ‘reciprocal’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.reciprocal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.reciprocal`.

Returns

Result of element-wise ‘reciprocal’ operation.

Return type

band_hic_matrix

 See also

`numpy.reciprocal`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.reciprocal()
```

3.1.5.60 bandhic.band_hic_matrix.remainder

`band_hic_matrix.remainder(self, other, *args, **kwargs)`

Perform element-wise ‘remainder’ operation with two inputs.

Parameters

- **self** (*band_hic_matrix*) – First input matrix.
- **other** (*band_hic_matrix* or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.remainder`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.remainder`.

Returns

Result of element-wise ‘remainder’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.remainder`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.remainder(other)
```

3.1.5.61 `bandhic.band_hic_matrix.right_shift`

`band_hic_matrix.right_shift(self, other, *args, **kwargs)`

Perform element-wise ‘right_shift’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.right_shift`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.right_shift`.

Returns

Result of element-wise ‘right_shift’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.right_shift`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.right_shift(other)
```

3.1.5.62 `bandhic.band_hic_matrix.rint`

`band_hic_matrix.rint(self, *args, **kwargs)`

Perform element-wise ‘rint’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy rint`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy rint`.

Returns

Result of element-wise ‘rint’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.rint`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.rint()
```

3.1.5.63 `bandhic.band_hic_matrix.sign`

`band_hic_matrix.sign(self, *args, **kwargs)`

Perform element-wise ‘sign’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sign`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sign`.

Returns

Result of element-wise ‘sign’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sign`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.sign()
```

3.1.5.64 `bandhic.band_hic_matrix.sin`

`band_hic_matrix.sin(self, *args, **kwargs)`

Perform element-wise ‘sin’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sin`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sin`.

Returns

Result of element-wise ‘sin’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sin`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.sin()
```

3.1.5.65 `bandhic.band_hic_matrix.sinh`

`band_hic_matrix.sinh(self, *args, **kwargs)`

Perform element-wise ‘sinh’ operation.

Parameters

- **self** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sinh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sinh`.

Returns

Result of element-wise ‘sinh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sinh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.sinh()
```

3.1.5.66 bandhic.band_hic_matrix.sqrt

`band_hic_matrix.sqrt(self, *args, **kwargs)`

Perform element-wise ‘sqrt’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.sqrt`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.sqrt`.

Returns

Result of element-wise ‘sqrt’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sqrt`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.sqrt()
```

3.1.5.67 bandhic.band_hic_matrix.square

`band_hic_matrix.square(self, *args, **kwargs)`

Perform element-wise ‘square’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.square`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.square`.

Returns

Result of element-wise ‘square’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.square`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.square()
```

3.1.5.68 bandhic.band_hic_matrix.subtract

`band_hic_matrix.subtract(self, other, *args, **kwargs)`

Perform element-wise ‘subtract’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.subtract`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.subtract`.

Returns

Result of element-wise ‘subtract’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.subtract`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.subtract(other)
```

3.1.5.69 bandhic.band_hic_matrix.tan

`band_hic_matrix.tan(self, *args, **kwargs)`

Perform element-wise ‘tan’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.tan`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.tan`.

Returns

Result of element-wise ‘tan’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.tan`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.tan()
```

3.1.5.70 bandhic.band_hic_matrix.tanh

`band_hic_matrix.tanh(self, *args, **kwargs)`

Perform element-wise ‘tanh’ operation.

Parameters

- **self** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.tanh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.tanh`.

Returns

Result of element-wise ‘tanh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.tanh`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = mat.tanh()
```

3.1.5.71 bandhic.band_hic_matrix.true_divide

`band_hic_matrix.true_divide(self, other, *args, **kwargs)`

Perform element-wise ‘true_divide’ operation with two inputs.

Parameters

- **self** (`band_hic_matrix`) – First input matrix.
- **other** (`band_hic_matrix` or *array-like*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.true_divide`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.true_divide`.

Returns

Result of element-wise ‘true_divide’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.true_divide`

Examples

```
>>> from bandhic import band_hic_matrix
>>> mat = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> other = mat.copy()
>>> result = mat.true_divide(other)
```

3.1.6 Data Type Conversion and Representation

This section includes methods for converting the matrix to different data types or representations, such as dense, sparse, or masked arrays.

<code>band_hic_matrix.todense()</code>	Convert the band matrix to a dense format.
<code>band_hic_matrix.tocoo([drop_zeros])</code>	Convert the matrix to COO format.
<code>band_hic_matrix.tocsr()</code>	Convert the matrix to CSR format.
<code>band_hic_matrix.copy()</code>	Deep copy the object.
<code>band_hic_matrix.astype(dtype[, copy])</code>	Cast data to new dtype.
<code>band_hic_matrix.__repr__()</code>	Return a string representation of the <code>band_hic_matrix</code> object.
<code>band_hic_matrix.__str__()</code>	Return a string representation of the <code>band_hic_matrix</code> object.
<code>band_hic_matrix.__array__([copy])</code>	Return the data as a NumPy array.
<code>band_hic_matrix.__array_priority__()</code>	Return the priority for array operations.

3.1.6.1 `bandhic.band_hic_matrix.todense`

`band_hic_matrix.todense()`

Convert the band matrix to a dense format.

Returns

The dense (square) matrix. Masked if mask is set.

Return type

ndarray or MaskedArray

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> dense = mat.todense()
>>> dense.shape
(3, 3)
```

3.1.6.2 `bandhic.band_hic_matrix.tocoo`

`band_hic_matrix.tocoo(drop_zeros=True)`

Convert the matrix to COO format.

Parameters

drop_zeros (*bool*, *optional*) – If True, zero entries will be dropped from the COO format. Default is True.

Returns

The matrix in scipy COO sparse format.

Return type

`coo_array`

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> coo = mat.tocoo()
>>> coo.shape
(3, 3)
```

3.1.6.3 `bandhic.band_hic_matrix.tocsr`

`band_hic_matrix.tocsr()`

Convert the matrix to CSR format.

Returns

The matrix in scipy CSR sparse format.

Return type

`csr_array`

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> csr = mat.tocsr()
>>> csr.shape
(3, 3)
```

3.1.6.4 `bandhic.band_hic_matrix.copy`

`band_hic_matrix.copy()`

Deep copy the object.

Returns

A deep copy.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat2 = mat.copy()
```

3.1.6.5 bandhic.band_hic_matrix.astype

`band_hic_matrix.astype(dtype, copy=False)`

Cast data to new dtype.

Parameters

- **type** (*data-type*) – Target dtype.
- **copy** (*bool*) – If True, the operation is performed in place. Default is False.
- **dtype** (*type*)

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat = mat.astype(np.float32)
```

3.1.6.6 bandhic.band_hic_matrix.__repr__

`band_hic_matrix.__repr__()`

Return a string representation of the `band_hic_matrix` object.

Returns

A string representation of the object.

Return type

str

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> repr(mat)
"band_hic_matrix(shape=(3, 3), diag_num=2, dtype=<class 'numpy.float64'>)"
```

3.1.6.7 bandhic.band_hic_matrix.__str__

`band_hic_matrix.__str__()`

Return a string representation of the `band_hic_matrix` object.

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
```

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```
>>> print(mat)
band_hic_matrix(shape=(3, 3), diag_num=2, dtype=<class 'numpy.float64'>)
```

3.1.6.8 bandhic.band_hic_matrix.__array__

`band_hic_matrix.__array__(copy=False)`

Return the data as a NumPy array.

Parameters

`copy` (*bool*, *optional*) – If True, returns a copy. Default is False.

Returns

Underlying band data array.

Return type

ndarray

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> arr = np.array(mat)
```

3.1.6.9 bandhic.band_hic_matrix.__array_priority__

`band_hic_matrix.__array_priority__()`

Return the priority for array operations.

Returns

Priority value.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> mat.__array_priority__()
100
```

3.1.7 Other Methods

This section includes additional methods that do not fall into the categories above but provide other useful operations for *band_hic_matrix*.

<code>band_hic_matrix.memory_usage()</code>	Compute memory usage of <i>band_hic_matrix</i> object.
<code>band_hic_matrix.__len__()</code>	Return the number of rows in the <i>band_hic_matrix</i> object.
<code>band_hic_matrix.__hash__()</code>	Return a hash value for the <i>band_hic_matrix</i> object.
<code>band_hic_matrix.__bool__()</code>	Truth value of the <i>band_hic_matrix</i> , following NumPy semantics.

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Table 6 – continued from previous page

<code>band_hic_matrix.dump(filename)</code>	Save the <code>band_hic_matrix</code> object to a file.
---	---

3.1.7.1 `bandhic.band_hic_matrix.memory_usage`

`band_hic_matrix.memory_usage()`

Compute memory usage of `band_hic_matrix` object.

Returns

Size in bytes.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat.memory_usage()
772
```

3.1.7.2 `bandhic.band_hic_matrix.__len__`

`band_hic_matrix.__len__()`

Return the number of rows in the `band_hic_matrix` object.

Returns

Number of rows.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> len(mat)
3
```

3.1.7.3 `bandhic.band_hic_matrix.__hash__`

`band_hic_matrix.__hash__()`

Return a hash value for the `band_hic_matrix` object.

Returns

Hash value.

Return type

int

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(3), diag_num=2)
>>> hashed=hash(mat)
```

3.1.7.4 bandhic.band_hic_matrix.__bool__

`band_hic_matrix.__bool__()`

Truth value of the `band_hic_matrix`, following NumPy semantics.

Returns

If the matrix has exactly one element (shape (1,1)), returns its truth value; otherwise raises a `ValueError`.

Return type

`bool`

Raises

ValueError – If the matrix contains more than one element.

3.1.7.5 bandhic.band_hic_matrix.dump

`band_hic_matrix.dump(filename)`

Save the `band_hic_matrix` object to a file.

Parameters

filename (*str*) – The name of the file to save the object to.

Return type

`None`

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((3,3), diag_num=2)
>>> mat.dump('myfile.npz')
```

3.2 Creation Functions

These functions create BandHiC matrices with specific properties.

<code>bandhic.ones(shape[, diag_num, dtype])</code>	Create a <code>band_hic_matrix</code> object filled with ones.
<code>bandhic.zeros(shape[, diag_num, dtype])</code>	Create a <code>band_hic_matrix</code> object filled with zeros.
<code>bandhic.eye(shape[, diag_num, dtype])</code>	Create a <code>band_hic_matrix</code> object filled as an identity matrix.
<code>bandhic.full(shape, fill_value[, diag_num, ...])</code>	Create a <code>band_hic_matrix</code> object filled with a specified value.
<code>bandhic.ones_like(other[, dtype])</code>	Create a <code>band_hic_matrix</code> object matching another matrix, filled with ones.
<code>bandhic.zeros_like(other[, dtype])</code>	Create a <code>band_hic_matrix</code> object matching another matrix, filled with zeros.
<code>bandhic.eye_like(other[, dtype])</code>	Create a <code>band_hic_matrix</code> object matching another matrix, filled as an identity matrix.
<code>bandhic.full_like(other, fill_value[, dtype])</code>	Create a <code>band_hic_matrix</code> object matching another matrix, filled with a specified value.

3.2.1 bandhic.ones

`bandhic.ones(shape, diag_num=1, dtype=<class 'numpy.float64'>)`

Create a `band_hic_matrix` object filled with ones.

Parameters

- **shape** (*tuple of int*) – Matrix shape as (bins, bins).
- **diag_num** (*int, optional*) – Number of diagonals to consider. Default is 1.
- **dtype** (*data-type, optional*) – The data type of the matrix. Default is `np.float64`.

Returns

A `band_hic_matrix` object with all entries filled with ones.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.ones((5, 5), diag_num=3)
>>> print(mat)
band_hic_matrix(shape=(5, 5), diag_num=3, dtype=<class 'numpy.float64'>)
```

3.2.2 bandhic.zeros

`bandhic.zeros(shape, diag_num=1, dtype=<class 'numpy.float64'>)`

Create a `band_hic_matrix` object filled with zeros.

Parameters

- **shape** (*tuple of int*) – Matrix shape as (bins, bins).
- **diag_num** (*int, optional*) – Number of diagonals to consider. Default is 1.
- **dtype** (*data-type, optional*) – The data type of the matrix. Default is `np.float64`.

Returns

A `band_hic_matrix` object with all entries filled with zeros.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.zeros((5, 5), diag_num=3)
>>> print(mat)
band_hic_matrix(shape=(5, 5), diag_num=3, dtype=<class 'numpy.float64'>)
```

3.2.3 bandhic.eye

`bandhic.eye(shape, diag_num=1, dtype=<class 'numpy.float64'>)`

Create a `band_hic_matrix` object filled as an identity matrix.

Parameters

- **shape** (*tuple of int*) – Matrix shape as (bins, bins).
- **diag_num** (*int, optional*) – Number of diagonals to consider. Default is 1.
- **dtype** (*data-type, optional*) – The data type of the matrix. Default is `np.float64`.

Returns

A `band_hic_matrix` object with ones on the main diagonal and zeros elsewhere.

Return type

band_hic_matrix

Examples

```
>>> mat = eye((5, 5), diag_num=3)
>>> print(mat)
band_hic_matrix(shape=(5, 5), diag_num=3, dtype=<class 'numpy.float64'>)
```

3.2.4 bandhic.full

`bandhic.full(shape, fill_value, diag_num=1, dtype=<class 'numpy.float64'>)`

Create a `band_hic_matrix` object filled with a specified value.

Parameters

- **shape** (*tuple of int*) – Matrix shape as (bins, bins).
- **fill_value** (*scalar*) – Value to fill the matrix with.
- **diag_num** (*int, optional*) – Number of diagonals to consider. Default is 1.
- **dtype** (*data-type, optional*) – The data type of the matrix. Default is `np.float64`.

Returns

A `band_hic_matrix` object with all entries filled with *fill_value*.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.full((5, 5), fill_value=7, diag_num=3)
>>> print(mat)
band_hic_matrix(shape=(5, 5), diag_num=3, dtype=<class 'numpy.float64'>)
```

3.2.5 bandhic.ones_like

`bandhic.ones_like(other, dtype=None)`

Create a `band_hic_matrix` object matching another matrix, filled with ones.

Parameters

other (*band_hic_matrix*) – Reference matrix.

Returns

A `band_hic_matrix` object matching *other*, filled with ones.

Return type

band_hic_matrix

Examples

```
>>> mat_ref = zeros((4, 4), diag_num=2)
>>> mat = ones_like(mat_ref)
>>> isinstance(mat, band_hic_matrix)
True
```

3.2.6 `bandhic.zeros_like`

`bandhic.zeros_like(other, dtype=None)`

Create a `band_hic_matrix` object matching another matrix, filled with zeros.

Parameters

other (`band_hic_matrix`) – Reference matrix.

Returns

A `band_hic_matrix` object matching *other*, filled with zeros.

Return type

band_hic_matrix

Examples

```
>>> mat_ref = zeros((4, 4), diag_num=2)
>>> mat = zeros_like(mat_ref)
>>> isinstance(mat, band_hic_matrix)
True
```

3.2.7 `bandhic.eye_like`

`bandhic.eye_like(other, dtype=None)`

Create a `band_hic_matrix` object matching another matrix, filled as an identity matrix.

Parameters

other (`band_hic_matrix`) – Reference matrix.

Returns

A `band_hic_matrix` object matching *other*, filled as an identity matrix.

Return type

band_hic_matrix

Examples

```
>>> mat_ref = zeros((4, 4), diag_num=2)
>>> mat = eye_like(mat_ref)
>>> isinstance(mat, band_hic_matrix)
True
```

3.2.8 bandhic.full_like

`bandhic.full_like(other, fill_value, dtype=None)`

Create a `band_hic_matrix` object matching another matrix, filled with a specified value.

Parameters

- **other** (`band_hic_matrix`) – Reference matrix.
- **fill_value** (`scalar`) – Value to fill the matrix.

Returns

A `band_hic_matrix` object matching *other*, filled with *fill_value*.

Return type

band_hic_matrix

Examples

```
>>> mat_ref = zeros((4, 4), diag_num=2)
>>> mat = full_like(mat_ref, fill_value=9)
>>> isinstance(mat, band_hic_matrix)
True
```

3.3 Input/Output Functions

These functions handle input and output operations for BandHiC matrices, including saving and loading from various formats.

<code>bandhic.save_npz(file_name, mat)</code>	Save a <code>band_hic_matrix</code> to a .npz file.
<code>bandhic.load_npz(file_name)</code>	Load a <code>band_hic_matrix</code> from a .npz file.
<code>bandhic.straw_chr(hic_file, chrom, ...[, ...])</code>	Read Hi-C data from a .hic file and return a <code>band_hic_matrix</code> .
<code>bandhic.straw_all_chrs(hic_file, resolution, ...)</code>	Read Hi-C data from a .hic file for all chromosomes and return a dictionary of <code>band_hic_matrix</code> objects.
<code>bandhic.cooler_chr(file_path, chrom, diag_num)</code>	Read Hi-C data from a .cool or .mcool file and return a <code>band_hic_matrix</code> .
<code>bandhic.cooler_all_chrs(file_path, diag_num)</code>	Read Hi-C data from a .cool or .mcool file for all chromosomes and return a dictionary of <code>band_hic_matrix</code> objects.
<code>bandhic.cooler_chr_all_cells(file_path, ...)</code>	Read Hi-C data from a .scool file for a specific chromosome and return a dictionary of <code>band_hic_matrix</code> objects for all cells.
<code>bandhic.cooler_all_cells_all_chrs(file_path, ...)</code>	Read Hi-C data from a .scool file for all cells and return a dictionary of dictionaries of <code>band_hic_matrix</code> objects.

3.3.1 bandhic.save_npz

`bandhic.save_npz(file_name, mat)`

Save a `band_hic_matrix` to a .npz file.

Parameters

- **file_name** (`str`) – Path to save the .npz file.
- **mat** (`band_hic_matrix`) – The `band_hic_matrix` object to save.

Return type

None

Examples

```
>>> import bandhic as bh
>>> mat = bh.band_hic_matrix(np.eye(5), diag_num=3)
>>> save_npz('./test/sample.npz', mat)
```

3.3.2 bandhic.load_npz

`bandhic.load_npz(file_name)`

Load a `band_hic_matrix` from a .npz file.

Parameters

file_name (*str*) – Path to the .npz file.

Returns

A `band_hic_matrix` object loaded from the file.

Return type

band_hic_matrix

Examples

```
>>> import bandhic as bh
>>> mat = bh.load_npz('./test/sample.npz')
>>> isinstance(mat, band_hic_matrix)
True
```

3.3.3 bandhic.straw_chr

`bandhic.straw_chr(hic_file, chrom, resolution, diag_num, data_type='observed', normalization='NONE', unit='BP')`

Read Hi-C data from a .hic file and return a `band_hic_matrix`.

Parameters

- **hic_file** (*str*) – Path to the .hic file. This file should be in the Hi-C format compatible with `hicstraw`. Local or remote paths are supported.
- **chrom** (*str*) – Chromosome name (e.g., 'chr1', 'chrX'). Short names like '1', 'X' are also accepted.
- **resolution** (*int*) – Resolution of the Hi-C data. Such as 10000 for 10kb resolution.
- **diag_num** (*int*) – Number of diagonals to consider.
- **data_type** (*str, optional*) – Type of data to read from the Hi-C file. Default is 'observed'. Other options include 'expected', 'balanced', etc. See `hicstraw`'s documentation for more details.
- **normalization** (*str, optional*) – Normalization method to apply. Default is 'NONE'. Other options include 'VC', 'VC_SQRT', 'KR', 'SCALE', etc. See `hicstraw` documentation for more details.
- **unit** (*str, optional*) – Unit of measurement for the Hi-C data. Default is 'BP' (base pairs). Other options include 'FRAG' (fragments), etc.

Return type

`band_hic_matrix`

 See also

`hicstraw` [documentation](#) Python interface to read Hi-C data files using *.hic* format.

Returns

A `band_hic_matrix` object containing the Hi-C data.

Return type

`band_hic_matrix`

Raises

ValueError – If the file cannot be parsed or parameters are invalid.

Parameters

- **hic_file** (*str*)
- **chrom** (*str*)
- **resolution** (*int*)
- **diag_num** (*int*)
- **data_type** (*str*)
- **normalization** (*str*)
- **unit** (*str*)

Examples

```
>>> import bandhic as bh
>>> mat = bh.straw_chr('/Users/wwb/Documents/workspace/BandHiC-Master/data/
↳ GSE130275_mESC_WT_combined_1.3B_microc.hic', 'chr1', resolution=10000, diag_
↳ num=200)
>>> isinstance(mat, band_hic_matrix)
True
```

3.3.4 bandhic.straw_all_chrs

`bandhic.straw_all_chrs(hic_file, resolution, diag_num, data_type='observed', normalization='NONE', unit='BP')`

Read Hi-C data from a *.hic* file for all chromosomes and return a dictionary of `band_hic_matrix` objects.

Parameters

- **hic_file** (*str*) – Path to the *.hic* file. This file should be in the Hi-C format compatible with `hicstraw`. Local or remote paths are supported.
- **resolution** (*int*) – Resolution of the Hi-C data. Such as 10000 for 10kb resolution.
- **diag_num** (*int*) – Number of diagonals to consider.
- **data_type** (*str, optional*) – Type of data to read from the Hi-C file. Default is 'observed'. Other options include 'expected', 'balanced', etc. See *hicstraw* documentation for more details.

- **normalization** (*str*, *optional*) – Normalization method to apply. Default is ‘NONE’. Other options include ‘VC’, ‘VC_SQRT’, ‘KR’, ‘SCALE’, etc. See *hicstraw* documentation for more details.
- **unit** (*str*, *optional*) – Unit of measurement for the Hi-C data. Default is ‘BP’ (base pairs). Other options include ‘FRAG’ (fragments), etc.

Returns

A dictionary mapping chromosome names to `band_hic_matrix` objects containing the Hi-C data.

Return type

`Dict[str, band_hic_matrix]`

Raises

ValueError – If the file cannot be parsed or parameters are invalid.

Examples

```
>>> import bandhic as bh
>>> mats = bh.straw_all_chrs('/Users/wwb/Documents/workspace/BandHiC-Master/data/
↳ GSE130275_mESC_WT_combined_1.3B_microc.hic', resolution=10000, diag_num=200)
>>> isinstance(mats['chr1'], band_hic_matrix)
True
```

3.3.5 bandhic.cooler_chr

`bandhic.cooler_chr(file_path, chrom, diag_num, cell_id=None, resolution=None, balance=True)`

Read Hi-C data from a .cool or .mcool file and return a `band_hic_matrix`.

Parameters

- **file_path** (*str*) – Path to the .cool, .mcool or .scool file.
- **chrom** (*str*) – Chromosome name.
- **diag_num** (*int*) – Number of diagonals to consider.
- **cell_id** (*str*, *optional*) – Cell ID for .scool files.
- **resolution** (*int*, *optional*) – Resolution of the Hi-C data.
- **balance** (*bool*, *optional*) – If True, use balanced data. Default is False. This parameter is specific to cooler files.

Returns

A `band_hic_matrix` object containing the Hi-C data.

Return type

band_hic_matrix

Raises

ValueError – If the cooler file is invalid or parameters are incorrect.

Examples

```
>>> import bandhic as bh
>>> mat = bh.cooler_chr('/Users/wwb/Documents/workspace/BandHiC-Master/data/yeast.
↳ 10kb.cool', 'chrI', resolution=10000, diag_num=10)
>>> isinstance(mat, band_hic_matrix)
True
```

➡ See also

[cooler documentation](#) Official documentation for the Cooler format and API usage.

3.3.6 bandhic.cooler_all_chrs

`bandhic.cooler_all_chrs(file_path, diag_num, resolution=None, cell_id=None, balance=True)`

Read Hi-C data from a .cool or .mcool file for all chromosomes and return a dictionary of band_hic_matrix objects.

Parameters

- **file_path** (*str*) – Path to the .cool, .mcool or .scool file.
- **diag_num** (*int*) – Number of diagonals to consider.
- **resolution** (*int*, *optional*) – Resolution of the Hi-C data.
- **cell_id** (*str*, *optional*) – Cell ID for .scool files.
- **balance** (*bool*, *optional*) – If True, use balanced data. Default is False. This parameter is specific to cooler files.

Returns

A dictionary mapping chromosome names to band_hic_matrix objects containing the Hi-C data.

Return type

Dict[str, *band_hic_matrix*]

Raises

ValueError – If the cooler file is invalid or parameters are incorrect.

Examples

```
>>> import bandhic as bh
>>> mats = bh.cooler_all_chrs('/Users/wwb/Documents/workspace/BandHiC-Master/data/
↳ yeast.10kb.cool', diag_num=10, resolution=10000)
>>> isinstance(mats['chrI'], band_hic_matrix)
True
```

3.3.7 bandhic.cooler_chr_all_cells

`bandhic.cooler_chr_all_cells(file_path, chrom, diag_num, balance=True)`

Read Hi-C data from a .scool file for a specific chromosome and return a dictionary of band_hic_matrix objects for all cells.

Parameters

- **file_path** (*str*) – Path to the .scool file.
- **chrom** (*str*) – Chromosome name.
- **diag_num** (*int*) – Number of diagonals to consider.
- **balance** (*bool*, *optional*) – If True, use balanced data. Default is False. This parameter is specific to cooler files.

Returns

A dictionary mapping cell IDs to band_hic_matrix objects for the specified chromosome.

Return type

Dict[str, *band_hic_matrix*]

Raises

ValueError – If the scool file is invalid or parameters are incorrect.

Examples

```
>>> import bandhic as bh
>>> mats = bh.cooler_chr_all_cells('/Users/wwb/Documents/workspace/BandHiC-Master/
↳ data/yeast.10kb.scool', 'chrI', diag_num=10, resolution=10000)
>>> isinstance(mats['cell1'], band_hic_matrix)
True
```

3.3.8 bandhic.cooler_all_cells_all_chrs

`bandhic.cooler_all_cells_all_chrs(file_path, diag_num, resolution=None)`

Read Hi-C data from a .scool file for all cells and return a dictionary of dictionaries of *band_hic_matrix* objects.

Parameters

- **file_path** (*str*) – Path to the .scool file.
- **diag_num** (*int*) – Number of diagonals to consider.
- **resolution** (*int*, *optional*) – Resolution of the Hi-C data.

Returns

A dictionary mapping cell IDs to dictionaries mapping chromosome names to *band_hic_matrix* objects.

Return type

Dict[str, Dict[str, *band_hic_matrix*]]

Raises

ValueError – If the scool file is invalid or parameters are incorrect.

Examples

```
>>> import bandhic as bh
>>> mats = bh.cooler_all_cells('/Users/wwb/Documents/workspace/BandHiC-Master/data/
↳ yeast.10kb.scool', diag_num=10, resolution=10000)
>>> isinstance(mats['cell1']['chrI'], band_hic_matrix)
True
```

3.4 Vectorized operations for BandHiC matrices

These functions perform element-wise operations on BandHiC matrices, allowing for efficient computation across the matrix elements. These functions are wrapped around numpy ufuncs to provide a consistent interface for element-wise operations.

<i>bandhic.absolute</i> (x, *args, **kwargs)	Perform element-wise 'absolute' operation.
<i>bandhic.add</i> (x1, x2, *args, **kwargs)	Perform element-wise 'add' operation with two inputs.
<i>bandhic.arccos</i> (x, *args, **kwargs)	Perform element-wise 'arccos' operation.

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<i>bandhic.arccosh</i> (x, *args, **kwargs)	Perform element-wise 'arccosh' operation.
<i>bandhic.arcsin</i> (x, *args, **kwargs)	Perform element-wise 'arcsin' operation.
<i>bandhic.arcsinh</i> (x, *args, **kwargs)	Perform element-wise 'arcsinh' operation.
<i>bandhic.arctan</i> (x, *args, **kwargs)	Perform element-wise 'arctan' operation.
<i>bandhic.arctan2</i> (x1, x2, *args, **kwargs)	Perform element-wise 'arctan2' operation with two inputs.
<i>bandhic.arctanh</i> (x, *args, **kwargs)	Perform element-wise 'arctanh' operation.
<i>bandhic.bitwise_and</i> (x1, x2, *args, **kwargs)	Perform element-wise 'bitwise_and' operation with two inputs.
<i>bandhic.bitwise_or</i> (x1, x2, *args, **kwargs)	Perform element-wise 'bitwise_or' operation with two inputs.
<i>bandhic.bitwise_xor</i> (x1, x2, *args, **kwargs)	Perform element-wise 'bitwise_xor' operation with two inputs.
<i>bandhic.cbrt</i> (x, *args, **kwargs)	Perform element-wise 'cbrt' operation.
<i>bandhic.conj</i> (x, *args, **kwargs)	Perform element-wise 'conj' operation.
<i>bandhic.conjugate</i> (x, *args, **kwargs)	Perform element-wise 'conjugate' operation.
<i>bandhic.cos</i> (x, *args, **kwargs)	Perform element-wise 'cos' operation.
<i>bandhic.cosh</i> (x, *args, **kwargs)	Perform element-wise 'cosh' operation.
<i>bandhic.deg2rad</i> (x, *args, **kwargs)	Perform element-wise 'deg2rad' operation.
<i>bandhic.degrees</i> (x, *args, **kwargs)	Perform element-wise 'degrees' operation.
<i>bandhic.divide</i> (x1, x2, *args, **kwargs)	Perform element-wise 'divide' operation with two inputs.
<i>bandhic.divmod</i> (x1, x2, *args, **kwargs)	Perform element-wise 'divmod' operation with two inputs.
<i>bandhic.equal</i> (x1, x2, *args, **kwargs)	Perform element-wise 'equal' operation with two inputs.
<i>bandhic.exp</i> (x, *args, **kwargs)	Perform element-wise 'exp' operation.
<i>bandhic.exp2</i> (x, *args, **kwargs)	Perform element-wise 'exp2' operation.
<i>bandhic.expm1</i> (x, *args, **kwargs)	Perform element-wise 'expm1' operation.
<i>bandhic.fabs</i> (x, *args, **kwargs)	Perform element-wise 'fabs' operation.
<i>bandhic.float_power</i> (x1, x2, *args, **kwargs)	Perform element-wise 'float_power' operation with two inputs.
<i>bandhic.floor_divide</i> (x1, x2, *args, **kwargs)	Perform element-wise 'floor_divide' operation with two inputs.
<i>bandhic.fmod</i> (x1, x2, *args, **kwargs)	Perform element-wise 'fmod' operation with two inputs.
<i>bandhic.gcd</i> (x1, x2, *args, **kwargs)	Perform element-wise 'gcd' operation with two inputs.
<i>bandhic.greater</i> (x1, x2, *args, **kwargs)	Perform element-wise 'greater' operation with two inputs.
<i>bandhic.greater_equal</i> (x1, x2, *args, **kwargs)	Perform element-wise 'greater_equal' operation with two inputs.
<i>bandhic.heaviside</i> (x1, x2, *args, **kwargs)	Perform element-wise 'heaviside' operation with two inputs.
<i>bandhic.hypot</i> (x1, x2, *args, **kwargs)	Perform element-wise 'hypot' operation with two inputs.
<i>bandhic.invert</i> (x, *args, **kwargs)	Perform element-wise 'invert' operation.
<i>bandhic.lcm</i> (x1, x2, *args, **kwargs)	Perform element-wise 'lcm' operation with two inputs.
<i>bandhic.left_shift</i> (x1, x2, *args, **kwargs)	Perform element-wise 'left_shift' operation with two inputs.
<i>bandhic.less</i> (x1, x2, *args, **kwargs)	Perform element-wise 'less' operation with two inputs.
<i>bandhic.less_equal</i> (x1, x2, *args, **kwargs)	Perform element-wise 'less_equal' operation with two inputs.
<i>bandhic.log</i> (x, *args, **kwargs)	Perform element-wise 'log' operation.
<i>bandhic.log1p</i> (x, *args, **kwargs)	Perform element-wise 'log1p' operation.
<i>bandhic.log2</i> (x, *args, **kwargs)	Perform element-wise 'log2' operation.
<i>bandhic.log10</i> (x, *args, **kwargs)	Perform element-wise 'log10' operation.

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<code>bandhic.logaddexp(x1, x2, *args, **kwargs)</code>	Perform element-wise 'logaddexp' operation with two inputs.
<code>bandhic.logaddexp2(x1, x2, *args, **kwargs)</code>	Perform element-wise 'logaddexp2' operation with two inputs.
<code>bandhic.logical_and(x1, x2, *args, **kwargs)</code>	Perform element-wise 'logical_and' operation with two inputs.
<code>bandhic.logical_or(x1, x2, *args, **kwargs)</code>	Perform element-wise 'logical_or' operation with two inputs.
<code>bandhic.logical_xor(x1, x2, *args, **kwargs)</code>	Perform element-wise 'logical_xor' operation with two inputs.
<code>bandhic.maximum(x1, x2, *args, **kwargs)</code>	Perform element-wise 'maximum' operation with two inputs.
<code>bandhic.minimum(x1, x2, *args, **kwargs)</code>	Perform element-wise 'minimum' operation with two inputs.
<code>bandhic.mod(x1, x2, *args, **kwargs)</code>	Perform element-wise 'mod' operation with two inputs.
<code>bandhic.multiply(x1, x2, *args, **kwargs)</code>	Perform element-wise 'multiply' operation with two inputs.
<code>bandhic.negative(x, *args, **kwargs)</code>	Perform element-wise 'negative' operation.
<code>bandhic.not_equal(x1, x2, *args, **kwargs)</code>	Perform element-wise 'not_equal' operation with two inputs.
<code>bandhic.positive(x, *args, **kwargs)</code>	Perform element-wise 'positive' operation.
<code>bandhic.power(x1, x2, *args, **kwargs)</code>	Perform element-wise 'power' operation with two inputs.
<code>bandhic.rad2deg(x, *args, **kwargs)</code>	Perform element-wise 'rad2deg' operation.
<code>bandhic.radians(x, *args, **kwargs)</code>	Perform element-wise 'radians' operation.
<code>bandhic.reciprocal(x, *args, **kwargs)</code>	Perform element-wise 'reciprocal' operation.
<code>bandhic.remainder(x1, x2, *args, **kwargs)</code>	Perform element-wise 'remainder' operation with two inputs.
<code>bandhic.right_shift(x1, x2, *args, **kwargs)</code>	Perform element-wise 'right_shift' operation with two inputs.
<code>bandhic rint(x, *args, **kwargs)</code>	Perform element-wise 'rint' operation.
<code>bandhic.sign(x, *args, **kwargs)</code>	Perform element-wise 'sign' operation.
<code>bandhic.sin(x, *args, **kwargs)</code>	Perform element-wise 'sin' operation.
<code>bandhic.sinh(x, *args, **kwargs)</code>	Perform element-wise 'sinh' operation.
<code>bandhic.sqrt(x, *args, **kwargs)</code>	Perform element-wise 'sqrt' operation.
<code>bandhic.square(x, *args, **kwargs)</code>	Perform element-wise 'square' operation.
<code>bandhic.subtract(x1, x2, *args, **kwargs)</code>	Perform element-wise 'subtract' operation with two inputs.
<code>bandhic.tan(x, *args, **kwargs)</code>	Perform element-wise 'tan' operation.
<code>bandhic.tanh(x, *args, **kwargs)</code>	Perform element-wise 'tanh' operation.
<code>bandhic.true_divide(x1, x2, *args, **kwargs)</code>	Perform element-wise 'true_divide' operation with two inputs.

3.4.1 bandhic.absolute

`bandhic.absolute(x, *args, **kwargs)`

Perform element-wise 'absolute' operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.absolute`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.absolute`.

Returns

Result of element-wise ‘absolute’ operation.

Return type

band_hic_matrix

 See also

`numpy.absolute`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.absolute(x1)
```

3.4.2 bandhic.add

`bandhic.add(x1, x2, *args, **kwargs)`

Perform element-wise ‘add’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.add`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.add`.

Returns

Result of element-wise ‘add’ operation.

Return type

band_hic_matrix

 See also

`numpy.add`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.add(x1, x2)
```

3.4.3 bandhic.arccos

`bandhic.arccos(x, *args, **kwargs)`

Perform element-wise ‘arccos’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arccos`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arccos`.

Returns

Result of element-wise ‘arccos’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arccos`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arccos(x1)
```

3.4.4 bandhic.arccosh

`bandhic.arccosh(x, *args, **kwargs)`

Perform element-wise ‘arccosh’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arccosh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arccosh`.

Returns

Result of element-wise ‘arccosh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arccosh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arccosh(x1)
```

3.4.5 bandhic.arcsin

`bandhic.arcsin(x, *args, **kwargs)`

Perform element-wise ‘arcsin’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arcsin`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arcsin`.

Returns

Result of element-wise ‘arcsin’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arcsin`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arcsin(x1)
```

3.4.6 bandhic.arcsinh

`bandhic.arcsinh(x, *args, **kwargs)`

Perform element-wise ‘arcsinh’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arcsinh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arcsinh`.

Returns

Result of element-wise ‘arcsinh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arcsinh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arcsinh(x1)
```

3.4.7 bandhic.arctan

`bandhic.arctan(x, *args, **kwargs)`

Perform element-wise ‘arctan’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arctan`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arctan`.

Returns

Result of element-wise ‘arctan’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arctan`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arctan(x1)
```

3.4.8 bandhic.arctan2

`bandhic.arctan2(x1, x2, *args, **kwargs)`

Perform element-wise ‘arctan2’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.arctan2`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.arctan2`.

Returns

Result of element-wise ‘arctan2’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arctan2`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.arctan2(x1, x2)
```

3.4.9 bandhic.arctanh

`bandhic.arctanh(x, *args, **kwargs)`

Perform element-wise ‘arctanh’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.arctanh`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.arctanh`.

Returns

Result of element-wise ‘arctanh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.arctanh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.arctanh(x1)
```

3.4.10 bandhic.bitwise_and

`bandhic.bitwise_and(x1, x2, *args, **kwargs)`

Perform element-wise ‘bitwise_and’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, `ndarray` or `scalar`) – First input matrix.
- **x2** (`band_hic_matrix`, `ndarray` or `scalar`) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.bitwise_and`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.bitwise_and`.

Returns

Result of element-wise ‘bitwise_and’ operation.

Return type

band_hic_matrix

See also

`numpy.bitwise_and`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.bitwise_and(x1, x2)
```

3.4.11 `bandhic.bitwise_or`

`bandhic.bitwise_or(x1, x2, *args, **kwargs)`

Perform element-wise ‘bitwise_or’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.bitwise_or`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.bitwise_or`.

Returns

Result of element-wise ‘bitwise_or’ operation.

Return type

band_hic_matrix

See also

`numpy.bitwise_or`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.bitwise_or(x1, x2)
```

3.4.12 `bandhic.bitwise_xor`

`bandhic.bitwise_xor(x1, x2, *args, **kwargs)`

Perform element-wise ‘bitwise_xor’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.bitwise_xor`.

- ****kwargs** (*dict*) – Keyword arguments for `numpy.bitwise_xor`.

Returns

Result of element-wise ‘bitwise_xor’ operation.

Return type

band_hic_matrix

 See also

`numpy.bitwise_xor`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.bitwise_xor(x1, x2)
```

3.4.13 bandhic.cbrt

`bandhic.cbrt`(*x*, **args*, ***kwargs*)

Perform element-wise ‘cbrt’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cbrt`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cbrt`.

Returns

Result of element-wise ‘cbrt’ operation.

Return type

band_hic_matrix

 See also

`numpy.cbrt`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.cbrt(x1)
```

3.4.14 bandhic.conj

`bandhic.conj`(*x*, **args*, ***kwargs*)

Perform element-wise ‘conj’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.conj`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.conj`.

Returns

Result of element-wise ‘conj’ operation.

Return type

band_hic_matrix

 See also

`numpy.conj`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.conj(x1)
```

3.4.15 bandhic.conjugate

`bandhic.conjugate(x, *args, **kwargs)`

Perform element-wise ‘conjugate’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.conjugate`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.conjugate`.

Returns

Result of element-wise ‘conjugate’ operation.

Return type

band_hic_matrix

 See also

`numpy.conjugate`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.conjugate(x1)
```

3.4.16 bandhic.cos

`bandhic.cos(x, *args, **kwargs)`

Perform element-wise ‘cos’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cos`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cos`.

Returns

Result of element-wise ‘cos’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.cos`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.cos(x1)
```

3.4.17 bandhic.cosh

`bandhic.cosh(x, *args, **kwargs)`

Perform element-wise ‘cosh’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.cosh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.cosh`.

Returns

Result of element-wise ‘cosh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.cosh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.cosh(x1)
```

3.4.18 bandhic.deg2rad

`bandhic.deg2rad(x, *args, **kwargs)`

Perform element-wise ‘deg2rad’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.deg2rad`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.deg2rad`.

Returns

Result of element-wise ‘deg2rad’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.deg2rad`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.deg2rad(x1)
```

3.4.19 bandhic.degrees

`bandhic.degrees(x, *args, **kwargs)`

Perform element-wise ‘degrees’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.degrees`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.degrees`.

Returns

Result of element-wise ‘degrees’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.degrees`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.degrees(x1)
```

3.4.20 bandhic.divide

`bandhic.divide(x1, x2, *args, **kwargs)`

Perform element-wise ‘divide’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.divide`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.divide`.

Returns

Result of element-wise ‘divide’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.divide`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.divide(x1, x2)
```

3.4.21 bandhic.divmod

`bandhic.divmod(x1, x2, *args, **kwargs)`

Perform element-wise ‘divmod’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.divmod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.divmod`.

Returns

Result of element-wise ‘divmod’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.divmod`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.divmod(x1, x2)
```

3.4.22 bandhic.equal

`bandhic.equal(x1, x2, *args, **kwargs)`

Perform element-wise ‘equal’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.equal`.

Returns

Result of element-wise ‘equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.equal`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.equal(x1, x2)
```

3.4.23 bandhic.exp

`bandhic.exp(x, *args, **kwargs)`

Perform element-wise ‘exp’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.exp`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.exp`.

Returns

Result of element-wise ‘exp’ operation.

Return type

band_hic_matrix

 See also

[numpy.exp](#)

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.exp(x1)
```

3.4.24 bandhic.exp2

`bandhic.exp2(x, *args, **kwargs)`

Perform element-wise ‘exp2’ operation.

Parameters

- **x** ([band_hic_matrix](#)) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.exp2`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.exp2`.

Returns

Result of element-wise ‘exp2’ operation.

Return type

band_hic_matrix

 See also

[numpy.exp2](#)

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.exp2(x1)
```

3.4.25 bandhic.expm1

`bandhic.expm1(x, *args, **kwargs)`

Perform element-wise ‘expm1’ operation.

Parameters

- **x** ([band_hic_matrix](#)) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.expm1`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.expm1`.

Returns

Result of element-wise ‘expm1’ operation.

Return type

band_hic_matrix

See also

`numpy.expm1`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.expm1(x1)
```

3.4.26 bandhic.fabs

`bandhic.fabs(x, *args, **kwargs)`

Perform element-wise ‘fabs’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.fabs`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.fabs`.

Returns

Result of element-wise ‘fabs’ operation.

Return type

band_hic_matrix

See also

`numpy.fabs`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.fabs(x1)
```

3.4.27 bandhic.float_power

`bandhic.float_power(x1, x2, *args, **kwargs)`

Perform element-wise ‘float_power’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, *ndarray or scalar*) – First input matrix.
- **x2** (`band_hic_matrix`, *ndarray or scalar*) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.float_power`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.float_power`.

Returns

Result of element-wise ‘float_power’ operation.

Return type

band_hic_matrix

↗ See also

`numpy.float_power`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.float_power(x1, x2)
```

3.4.28 bandhic.floor_divide

`bandhic.floor_divide(x1, x2, *args, **kwargs)`

Perform element-wise ‘floor_divide’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.floor_divide`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.floor_divide`.

Returns

Result of element-wise ‘floor_divide’ operation.

Return type

band_hic_matrix

↗ See also

`numpy.floor_divide`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.floor_divide(x1, x2)
```

3.4.29 bandhic.fmod

`bandhic.fmod(x1, x2, *args, **kwargs)`

Perform element-wise ‘fmod’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.

- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.fmod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.fmod`.

Returns

Result of element-wise ‘fmod’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.fmod`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.fmod(x1, x2)
```

3.4.30 bandhic.gcd

`bandhic.gcd(x1, x2, *args, **kwargs)`

Perform element-wise ‘gcd’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.gcd`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.gcd`.

Returns

Result of element-wise ‘gcd’ operation.

Return type

band_hic_matrix

➔ See also

`numpy.gcd`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.gcd(x1, x2)
```

3.4.31 bandhic.greater

`bandhic.greater(x1, x2, *args, **kwargs)`

Perform element-wise ‘greater’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.greater`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.greater`.

Returns

Result of element-wise ‘greater’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.greater`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.greater(x1, x2)
```

3.4.32 bandhic.greater_equal

`bandhic.greater_equal(x1, x2, *args, **kwargs)`

Perform element-wise ‘greater_equal’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.greater_equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.greater_equal`.

Returns

Result of element-wise ‘greater_equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.greater_equal`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.greater_equal(x1, x2)
```

3.4.33 bandhic.heaviside

`bandhic.heaviside(x1, x2, *args, **kwargs)`

Perform element-wise ‘heaviside’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.heaviside`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.heaviside`.

Returns

Result of element-wise ‘heaviside’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.heaviside`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.heaviside(x1, x2)
```

3.4.34 bandhic.hypot

`bandhic.hypot(x1, x2, *args, **kwargs)`

Perform element-wise ‘hypot’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.hypot`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.hypot`.

Returns

Result of element-wise ‘hypot’ operation.

Return type

band_hic_matrix

See also

`numpy.hypot`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.hypot(x1, x2)
```

3.4.35 `bandhic.invert`

`bandhic.invert(x, *args, **kwargs)`

Perform element-wise ‘invert’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.invert`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.invert`.

Returns

Result of element-wise ‘invert’ operation.

Return type

band_hic_matrix

See also

`numpy.invert`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.invert(x1)
```

3.4.36 `bandhic.lcm`

`bandhic.lcm(x1, x2, *args, **kwargs)`

Perform element-wise ‘lcm’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, *ndarray or scalar*) – First input matrix.
- **x2** (`band_hic_matrix`, *ndarray or scalar*) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.lcm`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.lcm`.

Returns

Result of element-wise ‘lcm’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.lcm`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.lcm(x1, x2)
```

3.4.37 bandhic.left_shift

`bandhic.left_shift(x1, x2, *args, **kwargs)`

Perform element-wise ‘left_shift’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.left_shift`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.left_shift`.

Returns

Result of element-wise ‘left_shift’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.left_shift`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.left_shift(x1, x2)
```

3.4.38 bandhic.less

`bandhic.less(x1, x2, *args, **kwargs)`

Perform element-wise ‘less’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.less`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.less`.

Returns

Result of element-wise ‘less’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.less`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.less(x1, x2)
```

3.4.39 bandhic.less_equal

`bandhic.less_equal(x1, x2, *args, **kwargs)`

Perform element-wise ‘less_equal’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.less_equal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.less_equal`.

Returns

Result of element-wise ‘less_equal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.less_equal`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.less_equal(x1, x2)
```

3.4.40 bandhic.log

`bandhic.log(x, *args, **kwargs)`

Perform element-wise ‘log’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log`.

Returns

Result of element-wise ‘log’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.log(x1)
```

3.4.41 bandhic.log1p

`bandhic.log1p(x, *args, **kwargs)`

Perform element-wise ‘log1p’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.log1p`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.log1p`.

Returns

Result of element-wise ‘log1p’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log1p`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.log1p(x1)
```

3.4.42 bandhic.log2

`bandhic.log2(x, *args, **kwargs)`

Perform element-wise ‘log2’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.log2`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.log2`.

Returns

Result of element-wise ‘log2’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log2`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.log2(x1)
```

3.4.43 bandhic.log10

`bandhic.log10(x, *args, **kwargs)`

Perform element-wise ‘log10’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.log10`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.log10`.

Returns

Result of element-wise ‘log10’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.log10`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.log10(x1)
```

3.4.44 bandhic.logaddexp

`bandhic.logaddexp(x1, x2, *args, **kwargs)`

Perform element-wise ‘logaddexp’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logaddexp`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logaddexp`.

Returns

Result of element-wise ‘logaddexp’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logaddexp`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.logaddexp(x1, x2)
```

3.4.45 bandhic.logaddexp2

`bandhic.logaddexp2(x1, x2, *args, **kwargs)`

Perform element-wise ‘logaddexp2’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logaddexp2`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logaddexp2`.

Returns

Result of element-wise ‘logaddexp2’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logaddexp2`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.logaddexp2(x1, x2)
```

3.4.46 bandhic.logical_and

`bandhic.logical_and(x1, x2, *args, **kwargs)`

Perform element-wise ‘logical_and’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_and`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_and`.

Returns

Result of element-wise ‘logical_and’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logical_and`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.logical_and(x1, x2)
```

3.4.47 bandhic.logical_or

`bandhic.logical_or(x1, x2, *args, **kwargs)`

Perform element-wise ‘logical_or’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_or`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_or`.

Returns

Result of element-wise ‘logical_or’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logical_or`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.logical_or(x1, x2)
```

3.4.48 `bandhic.logical_xor`

`bandhic.logical_xor(x1, x2, *args, **kwargs)`

Perform element-wise ‘logical_xor’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.logical_xor`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.logical_xor`.

Returns

Result of element-wise ‘logical_xor’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.logical_xor`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.logical_xor(x1, x2)
```

3.4.49 `bandhic.maximum`

`bandhic.maximum(x1, x2, *args, **kwargs)`

Perform element-wise ‘maximum’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.maximum`.

- ****kwargs** (*dict*) – Keyword arguments for `numpy.maximum`.

Returns

Result of element-wise ‘maximum’ operation.

Return type

band_hic_matrix

 See also

`numpy.maximum`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.maximum(x1, x2)
```

3.4.50 bandhic.minimum

`bandhic.minimum(x1, x2, *args, **kwargs)`

Perform element-wise ‘minimum’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix, ndarray or scalar*) – First input matrix.
- **x2** (*band_hic_matrix, ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.minimum`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.minimum`.

Returns

Result of element-wise ‘minimum’ operation.

Return type

band_hic_matrix

 See also

`numpy.minimum`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.minimum(x1, x2)
```

3.4.51 bandhic.mod

`bandhic.mod(x1, x2, *args, **kwargs)`

Perform element-wise ‘mod’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.mod`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.mod`.

Returns

Result of element-wise ‘mod’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.mod`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.mod(x1, x2)
```

3.4.52 bandhic.multiply

`bandhic.multiply(x1, x2, *args, **kwargs)`

Perform element-wise ‘multiply’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.multiply`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.multiply`.

Returns

Result of element-wise ‘multiply’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.multiply`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.multiply(x1, x2)
```

3.4.53 bandhic.negative

`bandhic.negative(x, *args, **kwargs)`

Perform element-wise ‘negative’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.negative`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.negative`.

Returns

Result of element-wise ‘negative’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.negative`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.negative(x1)
```

3.4.54 bandhic.not_equal

`bandhic.not_equal(x1, x2, *args, **kwargs)`

Perform element-wise ‘not_equal’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, `ndarray` or `scalar`) – First input matrix.
- **x2** (`band_hic_matrix`, `ndarray` or `scalar`) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.not_equal`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.not_equal`.

Returns

Result of element-wise ‘not_equal’ operation.

Return type

band_hic_matrix

See also

`numpy.not_equal`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.not_equal(x1, x2)
```

3.4.55 `bandhic.positive`

`bandhic.positive(x, *args, **kwargs)`

Perform element-wise ‘positive’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.positive`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.positive`.

Returns

Result of element-wise ‘positive’ operation.

Return type

band_hic_matrix

See also

`numpy.positive`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.positive(x1)
```

3.4.56 `bandhic.power`

`bandhic.power(x1, x2, *args, **kwargs)`

Perform element-wise ‘power’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, *ndarray or scalar*) – First input matrix.
- **x2** (`band_hic_matrix`, *ndarray or scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.power`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.power`.

Returns

Result of element-wise ‘power’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.power`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.power(x1, x2)
```

3.4.57 bandhic.rad2deg

`bandhic.rad2deg(x, *args, **kwargs)`

Perform element-wise ‘rad2deg’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.rad2deg`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.rad2deg`.

Returns

Result of element-wise ‘rad2deg’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.rad2deg`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.rad2deg(x1)
```

3.4.58 bandhic.radians

`bandhic.radians(x, *args, **kwargs)`

Perform element-wise ‘radians’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.radians`.

- ****kwargs** (*dict*) – Keyword arguments for `numpy.radians`.

Returns

Result of element-wise ‘radians’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.radians`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.radians(x1)
```

3.4.59 bandhic.reciprocal

`bandhic.reciprocal`(*x*, **args*, ***kwargs*)

Perform element-wise ‘reciprocal’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.reciprocal`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.reciprocal`.

Returns

Result of element-wise ‘reciprocal’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.reciprocal`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.reciprocal(x1)
```

3.4.60 bandhic.remainder

`bandhic.remainder`(*x1*, *x2*, **args*, ***kwargs*)

Perform element-wise ‘remainder’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.

- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.remainder`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.remainder`.

Returns

Result of element-wise ‘remainder’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.remainder`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.remainder(x1, x2)
```

3.4.61 bandhic.right_shift

`bandhic.right_shift(x1, x2, *args, **kwargs)`

Perform element-wise ‘right_shift’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.right_shift`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.right_shift`.

Returns

Result of element-wise ‘right_shift’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.right_shift`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.right_shift(x1, x2)
```

3.4.62 bandhic.rint

`bandhic.rint(x, *args, **kwargs)`

Perform element-wise ‘rint’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.rint`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.rint`.

Returns

Result of element-wise ‘rint’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.rint`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.rint(x1)
```

3.4.63 bandhic.sign

`bandhic.sign(x, *args, **kwargs)`

Perform element-wise ‘sign’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sign`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sign`.

Returns

Result of element-wise ‘sign’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sign`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.sign(x1)
```

3.4.64 bandhic.sin

`bandhic.sin(x, *args, **kwargs)`

Perform element-wise ‘sin’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sin`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sin`.

Returns

Result of element-wise ‘sin’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sin`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.sin(x1)
```

3.4.65 bandhic.sinh

`bandhic.sinh(x, *args, **kwargs)`

Perform element-wise ‘sinh’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sinh`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sinh`.

Returns

Result of element-wise ‘sinh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sinh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.sinh(x1)
```

3.4.66 bandhic.sqrt

`bandhic.sqrt(x, *args, **kwargs)`

Perform element-wise ‘sqrt’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.sqrt`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.sqrt`.

Returns

Result of element-wise ‘sqrt’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.sqrt`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.sqrt(x1)
```

3.4.67 bandhic.square

`bandhic.square(x, *args, **kwargs)`

Perform element-wise ‘square’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.square`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.square`.

Returns

Result of element-wise ‘square’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.square`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.square(x1)
```

3.4.68 bandhic.subtract

`bandhic.subtract(x1, x2, *args, **kwargs)`

Perform element-wise ‘subtract’ operation with two inputs.

Parameters

- **x1** (*band_hic_matrix*, *ndarray* or *scalar*) – First input matrix.
- **x2** (*band_hic_matrix*, *ndarray* or *scalar*) – Second input for the operation.
- ***args** (*tuple*) – Additional positional arguments for `numpy.subtract`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.subtract`.

Returns

Result of element-wise ‘subtract’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.subtract`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.subtract(x1, x2)
```

3.4.69 bandhic.tan

`bandhic.tan(x, *args, **kwargs)`

Perform element-wise ‘tan’ operation.

Parameters

- **x** (*band_hic_matrix*) – Input matrix.
- ***args** (*tuple*) – Additional positional arguments for `numpy.tan`.
- ****kwargs** (*dict*) – Keyword arguments for `numpy.tan`.

Returns

Result of element-wise ‘tan’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.tan`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.tanh(x1)
```

3.4.70 bandhic.tanh

`bandhic.tanh(x, *args, **kwargs)`

Perform element-wise ‘tanh’ operation.

Parameters

- **x** (`band_hic_matrix`) – Input matrix.
- ***args** (`tuple`) – Additional positional arguments for `numpy.tanh`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.tanh`.

Returns

Result of element-wise ‘tanh’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.tanh`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> result = bh.tanh(x1)
```

3.4.71 bandhic.true_divide

`bandhic.true_divide(x1, x2, *args, **kwargs)`

Perform element-wise ‘true_divide’ operation with two inputs.

Parameters

- **x1** (`band_hic_matrix`, `ndarray` or `scalar`) – First input matrix.
- **x2** (`band_hic_matrix`, `ndarray` or `scalar`) – Second input for the operation.
- ***args** (`tuple`) – Additional positional arguments for `numpy.true_divide`.
- ****kwargs** (`dict`) – Keyword arguments for `numpy.true_divide`.

Returns

Result of element-wise ‘true_divide’ operation.

Return type

band_hic_matrix

➡ See also

`numpy.true_divide`

Examples

```
>>> import bandhic as bh
>>> x1 = band_hic_matrix(np.eye(3), diag_num=2, dtype=int)
>>> x2 = x1.copy()
>>> result = bh.true_divide(x1, x2)
```

3.5 Data Reduction and Normalization

These functions perform data reduction and normalization on BandHiC matrices, allowing for efficient analysis of Hi-C data. These functions are wrapped around `band_hic_matrix` methods to provide a consistent interface for data reduction and normalization.

<code>bandhic.min(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.min</code>
<code>bandhic.max(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.max</code>
<code>bandhic.sum(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.sum</code>
<code>bandhic.mean(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.mean</code>
<code>bandhic.std(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.std</code>
<code>bandhic.var(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.var</code>
<code>bandhic.prod(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.prod</code>
<code>bandhic.ptp(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.ptp</code>
<code>bandhic.all(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.all</code>
<code>bandhic.any(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.any</code>
<code>bandhic.clip(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.clip</code>
<code>bandhic.normalize(*args, **kwargs)</code>	Wrapped from <code>band_hic_matrix.normalize</code>

3.5.1 bandhic.min

`bandhic.min(*args, **kwargs)`

Wrapped from `band_hic_matrix.min`

Refer to the documentation: `bandhic.band_hic_matrix.min()`

3.5.2 bandhic.max

`bandhic.max(*args, **kwargs)`

Wrapped from `band_hic_matrix.max`

Refer to the documentation: `bandhic.band_hic_matrix.max()`

3.5.3 bandhic.sum

`bandhic.sum(*args, **kwargs)`

Wrapped from `band_hic_matrix.sum`

Refer to the documentation: `bandhic.band_hic_matrix.sum()`

3.5.4 bandhic.mean

`bandhic.mean(*args, **kwargs)`

Wrapped from *band_hic_matrix.mean*

Refer to the documentation: [*bandhic.band_hic_matrix.mean\(\)*](#)

3.5.5 bandhic.std

`bandhic.std(*args, **kwargs)`

Wrapped from *band_hic_matrix.std*

Refer to the documentation: [*bandhic.band_hic_matrix.std\(\)*](#)

3.5.6 bandhic.var

`bandhic.var(*args, **kwargs)`

Wrapped from *band_hic_matrix.var*

Refer to the documentation: [*bandhic.band_hic_matrix.var\(\)*](#)

3.5.7 bandhic.prod

`bandhic.prod(*args, **kwargs)`

Wrapped from *band_hic_matrix.prod*

Refer to the documentation: [*bandhic.band_hic_matrix.prod\(\)*](#)

3.5.8 bandhic.ptp

`bandhic.ptp(*args, **kwargs)`

Wrapped from *band_hic_matrix.ptp*

Refer to the documentation: [*bandhic.band_hic_matrix.ptp\(\)*](#)

3.5.9 bandhic.all

`bandhic.all(*args, **kwargs)`

Wrapped from *band_hic_matrix.all*

Refer to the documentation: [*bandhic.band_hic_matrix.all\(\)*](#)

3.5.10 bandhic.any

`bandhic.any(*args, **kwargs)`

Wrapped from *band_hic_matrix.any*

Refer to the documentation: [*bandhic.band_hic_matrix.any\(\)*](#)

3.5.11 bandhic.clip

`bandhic.clip(*args, **kwargs)`

Wrapped from *band_hic_matrix.clip*

Refer to the documentation: [*bandhic.band_hic_matrix.clip\(\)*](#)

3.5.12 bandhic.normalize

`bandhic.normalize(*args, **kwargs)`

Wrapped from `band_hic_matrix.normalize`

Refer to the documentation: `bandhic.band_hic_matrix.normalize()`

3.6 Other Functions

These functions provide additional utilities for BandHiC matrices, including matrix comparison, bias computation, and TAD calling.

<code>bandhic.matrix_equal(a, b)</code>	Check if two <code>band_hic_matrix</code> objects are equal.
<code>bandhic.assert_band_matrix_equal(a, b)</code>	Assert that two <code>band_hic_matrix</code> objects are equal.
<code>bandhic.compute_bin_bias(hic_coo[, verbose])</code>	Compute bias values for Hi-C contact matrices using the Knight-Ruiz normalization algorithm.
<code>bandhic.call_tad(hic_matrix, resolution, ...)</code>	Detect TADs from a Hi-C matrix using the TopDom algorithm.
<code>bandhic.topdom(hic_matrix, bins, window_size)</code>	Detect TADs using TopDom algorithm.

3.6.1 bandhic.matrix_equal

`bandhic.matrix_equal(a, b)`

Check if two `band_hic_matrix` objects are equal.

Parameters

- **a** (`band_hic_matrix`) – First `band_hic_matrix` object.
- **b** (`band_hic_matrix`) – Second `band_hic_matrix` object.

Returns

True if the matrices are equal, False otherwise.

Return type

bool

3.6.2 bandhic.assert_band_matrix_equal

`bandhic.assert_band_matrix_equal(a, b)`

Assert that two `band_hic_matrix` objects are equal.

Parameters

- **a** (`band_hic_matrix`) – First `band_hic_matrix` object.
- **b** (`band_hic_matrix`) – Second `band_hic_matrix` object.

Raises

AssertionError – If the two matrices are not equal.

Return type

bool

3.6.3 bandhic.compute_bin_bias

`bandhic.compute_bin_bias(hic_coo, verbose=False)`

Compute bias values for Hi-C contact matrices using the Knight-Ruiz normalization algorithm.

Parameters

- **hic_coo** (*scipy.sparse.coo_array*) – A sparse COO matrix representing Hi-C contact data.
- **verbose** (*bool, optional*) – If True, print detailed information during processing. Default is False.

Returns

- **bias** (*numpy.ndarray*) – A 1D array containing the bias values for each bin in the Hi-C matrix.
- **is_valid** (*bool*) – A boolean indicating whether the bias vector is valid (mean and median within typical range).

Notes

This function removes a specified percentage of the most sparse bins from the Hi-C matrix before computing the bias values. The Knight-Ruiz normalization algorithm is applied to the modified matrix to compute the bias. The function iteratively removes bins with low interaction counts until a valid bias vector is obtained. The bias vector is expected to have a mean and median close to 1, indicating balanced interaction frequencies across bins.

3.6.4 bandhic.call_tad

`bandhic.call_tad(hic_matrix, resolution, chrom_short, window_size_bp=200000, min_TAD_size=None, stat_filter=True, verbose=False)`

Detect TADs from a Hi-C matrix using the TopDom algorithm. Different from the *TopDom* function, this function accepts a Hi-C matrix in either *band_hic_matrix* format or as a dense numpy array. This function is designed to be used with the BandHiC package and provides a convenient interface for TAD detection.

Parameters

- **hic_matrix** (*band_hic_matrix* or *np.ndarray*) – The Hi-C matrix, either as a *band_hic_matrix* object or a dense numpy array.
- **resolution** (*int*) – The resolution of the Hi-C matrix.
- **chrom_short** (*str*) – The chromosome name without 'chr' prefix.
- **window_size_bp** (*int, optional*) – The size of the window to consider for detecting TADs, in base pairs. Default is 200000.
- **min_TAD_size** (*int, optional*) – The minimum size of a TAD to be considered valid, in base pairs. Default is None.
- **stat_filter** (*bool, optional*) – Whether to apply statistical filtering to remove false positives. Default is True.
- **verbose** (*bool, optional*) – Whether to print detailed information during processing. Default is False.

Returns

- **domains** (*pd.DataFrame*) – DataFrame containing detected TADs with columns 'chr', 'from.id', 'from.coord', 'to.id', 'to.coord', 'tag'.
- **bins** (*pd.DataFrame*) – DataFrame containing bin information columns 'id', 'chr', 'from.coord', 'to.coord', 'local.ext', 'mean.cf', 'pvalue'.

3.6.5 bandhic.topdom

`bandhic.topdom(hic_matrix, bins, window_size, stat_filter=True, verbose=False)`

Detect TADs using TopDom algorithm.

Parameters

- **hic_matrix** (`band_hic_matrix` or `np.ndarray`) – The Hi-C matrix, either as a `band_hic_matrix` object or a dense numpy array.
- **bins** (`pd.DataFrame`) – DataFrame containing bin information with columns ‘chr’, ‘from.coord’, ‘to.coord’.
- **window_size** (`int`) – Size of the window to consider for detecting TADs.
- **stat_filter** (`bool`, *optional*) – Whether to apply statistical filtering to remove false positives. Default is True.
- **verbose** (`bool`, *optional*) – Whether to print detailed information during processing. Default is False.

Returns

- **domains** (`pd.DataFrame`) – DataFrame containing detected TADs with columns ‘chr’, ‘from.id’, ‘from.coord’, ‘to.id’, ‘to.coord’, ‘tag’.
- **bins** (`pd.DataFrame`) – Updated DataFrame containing bin information with additional columns ‘local.ext’, ‘mean.cf’, ‘pvalue’.

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